Cubic twists of Legendre moments: cohomological descent and Frobenius trace formulas

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Abstract

We construct the parabolic cohomology representations attached to cubic twists of the second and fourth moments of the Legendre family. For the fourth moment, an explicit arithmetic involution gives two rational four-dimensional descents related by the quadratic character of $\mathbb{Q}(\sqrt{-3})$. We prove their alternating pairing and derive all good-prime Euler factors. Deligne's weight theorem gives good-prime purity of weight five. An explicit auxiliary-level construction realizes both descents by rational Chow projectors on a smooth projective fivefold; their Hodge numbers are $h^{5,0} = h^{0,5} = 2$, with no other Hodge types. At inert primes a norm-one-torus calculation determines the trace, the boundary contribution and the sign matching the earlier finite-sum recipe. The cubic second-moment modular comparison uses explicit finite-check hypotheses and the supplied bounded exact certificate for the displayed rows and Sturm comparison. We compute inertia invariants and Swan conductors at both bad primes. The literal plus descent has conductor 432, and its quadratic twist has conductor 1296. A local-monodromy and Hodge–Tate argument proves strong absolute irreducibility for coefficient primes $\ell \neq 3$. We retain the corrected hypergeometric normalization and distinguish these written local proofs from the remaining numerical special-value observations. The underlying cover is compared explicitly with existing literature; no general BSD consequence or literature-priority certification is asserted.

Keywords: Legendre family; character sums; Scholl representations; local monodromy; Artin conductors; parabolic cohomology.

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Contents

1 Introduction and statements 3

2 Relation to earlier work 4

3 Geometry and trace formulas over K 5

4 The cubic second moment and its modular comparison 9

5 Arithmetic involution, descent and polarization 12

6 Rational Scholl realization and Hodge numbers 16

7 Inert Frobenius traces and good Euler factors 20

8 Local inertia and the Euler factors at 3 23

9 The local factor and conductor exponent at 2 27

10 Conductor products and coefficient-prime scope 38

11 Strong irreducibility from local monodromy 39

12 Current claims and their dependencies 41

13 Numerical evidence, corrections and open questions 41

A Known quadratic trace identities 43

B The corrected hypergeometric normalization 44

C An analytic period substitution and the row-sign correction 45

1 Introduction and statements

Character-twisted moments of the Legendre family give concrete finite sums whose cohomological interpretation depends on boundary terms and arithmetic normalizations. This paper treats cubic characters and the second and fourth moments. Its central purpose is an explicit construction of the associated representations and their Frobenius formulas, including the inert-prime sign. The consolidated local arguments now also determine the two bad-prime factors and conductor exponents. The exact finite-sum prescription underlying the numerical degree-four recipe in the earlier preprint [1] is thereby identified at good primes with a specified representation.

1.1 Notation and Frobenius convention

Put $K = \mathbb{Q}(\zeta_3)$, $E = \mathbb{Q}_2(\zeta_3)$ and $U = \mathbb{P}^1 \setminus \{0, 1, \infty\}$. Let $\pi : \mathcal{E} \to U$ be the smooth projective Legendre family

$$\mathcal{E}_\lambda : y^2 = t(t-1)(t-\lambda), \qquad \mathcal{V} = R^1\pi_* E.$$

Let $\mathcal{L}_\rho$ be either nontrivial character summand of the actual arithmetic cover $z^3 = \lambda$. No additional constant Galois twist is allowed in this definition. For an integer $r \in \{2, 4\}$ set

$$H_r = H^1_!(U_{\overline{K}}, \mathrm{Sym}^r \mathcal{V} \otimes \mathcal{L}_\rho), \qquad H^1_! = \mathrm{im}(H^1_c \longrightarrow H^1), \qquad W_2 = H_2, \quad V_4 = H_4.$$

Frobenius always means *geometric Frobenius on actual cohomology.* Write τ for the character of $\mathbb{Q}_2(-1)$, so $\tau(\mathrm{Frob}_\mathfrak{p}) = N\mathfrak{p}$ and $\tau = \chi_{\mathrm{cyc}}^{-1}$ in the usual convention on $\mathbb{Q}_2(1)$. The modular comparison uses the cohomological dual of the conventional arithmetic-Frobenius Deligne representation.

For a good residue field of cardinality q and $\lambda \neq 0, 1$, put

$$a_q(\lambda) = q + 1 - \#\mathcal{E}_\lambda(\mathbb{F}_q), \qquad M_{3,j}(q) = \sum_{\lambda \in \mathbb{F}_q \setminus \{0,1\}} \rho(\lambda) a_q(\lambda)^j.$$

Here ρ is the trace character of the specified Kummer summand. When taking powers of one Frobenius, lift it by the finite-field norm. At an inert rational prime of K, $\mathbb{F}_{p^2}$ is the initial residue field; no nontrivial cubic character of $\mathbb{F}_p^\times$ is being lifted. Define

$$v(q) = -M_{3,4}(q) + 3qM_{3,2}(q) + q^2 - 1.$$

Theorem 1.1 (Summary of the fourth-moment results)**.** *The representation V_4 has dimension four over E and is unramified outside the primes over $2, 3$. At every good prime of K, $\mathrm{Tr}(\mathrm{Frob}_\mathfrak{p} \mid V_4) = v(N\mathfrak{p})$. There are explicit four-dimensional $\mathbb{Q}_2$-representations W_4^+ and W_4^- of $G_\mathbb{Q}$, unramified outside $2, 3$, whose restrictions to G_K after scalar extension are V_4, with*

$$W_4^- \simeq W_4^+ \otimes \chi_{-3}.$$

Both have a perfect alternating pairing with values in $\mathbb{Q}_2(-5)$ and are pure of weight five at every rational prime $p \geq 5$. Each is the 2-adic realization of a rational Chow motive cut out on a smooth projective fivefold, with Hodge type $(5,0)^2 + (0,5)^2$; see Theorem 6.1. For $p \geq 5$, $p \equiv 2 \pmod 3$, choose $d \in \mathbb{F}_p$ nonsquare, $r^2 = d$ in $\mathbb{F}_{p^2}$, and a primitive cubic character ρ there. Put

$$E'_x : Y^2 = X(X^2 - 2xX + x^2 - d), \quad a'_p(x) = p + 1 - \#E'_x(\mathbb{F}_p),$$

$$S_p = \sum_{x \in \mathbb{F}_p} \rho(x + r)(a'_p(x)^4 - 3p\, a'_p(x)^2 + p^2).$$

Then $S_p \in \mathbb{Z}$ is independent of these choices and

$$\mathrm{Tr}(\mathrm{Frob}_p \mid W_4^+) = -S_p - 1, \qquad \mathrm{Tr}(\mathrm{Frob}_p \mid W_4^-) = S_p + 1.$$

For W_4^-, set $c_1(p) = v(p)$ at split primes and $c_1(p) = S_p + 1$ at inert primes, and set $c_2(p) = (c_1(p)^2 - v(p^2))/2$ with the character convention above. Then

$$\det(1 - \mathrm{Frob}_p T \mid W_4^-) = 1 - c_1(p)T + c_2(p)T^2 - p^5 c_1(p)T^3 + p^{10}T^4.$$

Theorem 1.2 (Local data and strong irreducibility)**.** *Let $W_{4,\ell}^{\pm}$ be the realizations defined by the same rational projectors as $W_4^{\pm}$ in Theorem 6.1. At 2, for every $\ell \neq 2$, both have zero inertia invariants and Swan conductor zero. At 3, for every $\ell \neq 3$, the plus descent has a one-dimensional invariant space with geometric Frobenius 3, whereas the minus descent has no invariants; both Swan conductors vanish. Consequently the local Euler polynomials and Artin exponents are*

	$P_2(T)$	a_2	$P_3(T)$	a_3	N
W_4^+	1	4	$1-3T$	3	432
W_4^-	1	4	1	4	1296

Here N is the product of the local conductor powers, using a coefficient prime different from the residue characteristic for each local calculation. For every $\ell \neq 2, 3$, it is the prime-to-ℓ Artin conductor of $W_{4,\ell}^{\pm}$. The displayed local data are independent of ℓ within their stated ranges. For every $\ell \neq 3$, both realizations are absolutely irreducible and remain so after restriction to any finite number-field extension of $\mathbb{Q}$.

The local assertions are proved in Sections 8 and 9, the conductor deduction in Section 10, and strong irreducibility in Section 11. The last assertion includes the original 2-adic realizations; no irreducibility claim at coefficient prime 3 is made here.

The construction, pairing and inert calculation are proved in Sections 3, 5, 6 and 7. The cubic second-moment comparison is treated separately in Section 4; its modular conclusion uses the explicit exact finite checks stated and independently recomputed there. These rank-two and rank-four objects must not be conflated.

1.2 Scope of this revision

This consolidated research draft continues the version-2 preprint [26], with its earlier predecessor [1] retained for provenance. It brings the local-at-3, dyadic tameness, dyadic invariant, and strong-irreducibility arguments into the same manuscript. The earlier blanket open obligations for these local data are superseded by Theorem 1.2. Section 12 records the current proof dependencies and unresolved questions. This draft has internal AI-assisted review; external specialist review remains outstanding. It does not certify the archived floating-point tables or promote their functional-equation sign and special values to theorems.

The supplementary corrections retain known trace-formula consequences, the Jacobi-sum normalization, and an analytic period substitution. Those statements are explicitly distinguished from the main construction and from numerical special-value observations. No classical elliptic BSD rank or leading-term theorem is asserted.

2 Relation to earlier work

The cubic cover used here belongs to an established family. Kazalicki–Scholl [13, Sections 7–8] describe the genus-zero groups $\Phi_0(N)$ and the elliptic family $Y^2 = X(X+1)(X+t^N)$. For $N = 3$ and $t = z$, substituting $x = -X$ in our equation gives

$$y^2 = -X(X+1)(X+z^3).$$

Thus the two families differ by the constant quadratic twist -1. The corresponding Sym^4 coefficient systems are isomorphic over $\mathbb{Q}$, since the fourth power of the twisting character is trivial. Their analyzed weight-three representation uses R^1; it does not by this comparison identify the fourth-symmetric-power constituent studied here. Neither the underlying family nor its cubic-cover geometry is claimed as new.

Scholl's construction places parabolic cohomology of symmetric powers in the general theory of motives attached to noncongruence cusp forms; see [14, Introduction]. More precise projector

references are [22, Sections 1.1–1.2] and [23, Sections 2.2–2.8 and 4.2–4.4]. Our local calculations use the standard etale-cohomological framework. Section 6 applies these established constructions with an explicit full-level cover and projectors matching the literal normalized deck and inversion operators. The degree-four representations treated by Atkin–Li–Liu–Long [15] provide further relevant context. Their automorphy theorem requires quaternion multiplication satisfying specified compatibility conditions. We have not established those conditions for W_4^+ or W_4^- [15, Definition 3.1.1 and Theorem 4.3.1]. In fact $J = A - A^2$ exchanges the two spaces, while I is scalar on each; these operators are not a quaternion-multiplication pair on either four-dimensional summand.

The auxiliary moment identities also have substantial precedents. The quadratic-twist corollaries use the Legendre Hecke trace formulas recorded in [16, Theorem 2.3], with their attribution to Ahlgren and Ahlgren–Ono [18, 17]. The weight-four forms used in the second-moment comparisons occur in Long–Tu–Yui–Zudilin [12, Theorem 2 and Table 1]; those modular forms and the associated hypergeometric modularity are not discoveries of this note.

There is also an exact parameter overlap with Allen–Grove–Long–Tu [19]. The datum in our cubic second-moment normalization,

$$\left\{\begin{matrix}\frac{1}{2},\frac{1}{2},\frac{1}{3},\frac{1}{3}\\ 1,1,\frac{5}{6},\frac{5}{6}\end{matrix}\right\},$$

is their $\mathrm{HD}_4(j/12)$ at $j = 4$, defined in the published article's introduction immediately before equation (1.1). (The corresponding definition is equation (1.2) in arXiv v1.) Their Theorem 1.4 concerns its specialization at argument -1; our normalization identity uses argument 1 and an explicit squared Jacobi-sum factor. Equality of the parameter lists alone does not identify these specializations or identify either with $W_4^{\pm}$.

The results established here specify the arithmetic coefficient sheaf, construct its two rational descents, and derive their good-prime traces, including the inert-prime boundary sign. These statements concern the precisely normalized objects defined in the paper. Their exact priority has not been established by the literature comparison, and no priority claim is made for the constituent or trace formula. The conductor calculation in Sections 8–10 is an independent local argument, not a consequence of this comparison. No automorphic identification or classical elliptic BSD consequence follows from the literature overlap.

3 Geometry and trace formulas over K

Fix compatible primitive cube roots in $K = \mathbb{Q}(\zeta_3)$ and $E = \mathbb{Q}_2(\zeta_3)$, and put $\mathcal{O} = \mathbb{Z}_2[\zeta_3]$, $S = \operatorname{Spec}\mathcal{O}_K[1/6]$, and $U = \mathbb{P}^1_S \setminus \{0,1,\infty\}$. Let $\pi : \mathcal{E} \to U$ be the smooth projective Legendre elliptic scheme, with its zero section and affine equation

$$\mathcal{E}_\lambda : \quad y^2 = x(x-1)(x-\lambda), \qquad \mathcal{V} = R^1\pi_* E.$$

The cubic torsor $f : T \to U$, $z^3 = \lambda$, has deck transformation $a(z) = \zeta_3 z$. We use pullback as the deck action. Let $\mathcal{L}_\rho$ be the ζ_3-eigensheaf of a^* in f_*E. This definition fixes the arithmetic normalization; in particular no constant Galois twist is inserted. For $r \in \{2,4\}$ set

$$\mathcal{G}_r = \operatorname{Sym}^r \mathcal{V} \otimes \mathcal{L}_\rho, \qquad H_r = H^1_!(U_{\bar K}, \mathcal{G}_r) := \operatorname{im}(H^1_c(U_{\bar K}, \mathcal{G}_r) \longrightarrow H^1(U_{\bar K}, \mathcal{G}_r)).$$

Thus $W_2 = H_2$ and $V_4 = H_4$. The subscript records the symmetric power, and is distinct from the order three of the Kummer character.

All Frobenius operators below are geometric. Write τ for the character of $E(-1)$, so $\tau(\mathrm{Frob}_\mathfrak{p}) = N\mathfrak{p}$. For a prime $\mathfrak{p} \nmid 6$, put $q = N\mathfrak{p}$ and let $\rho_q : \mathbb{F}_q^\times \to E^\times$ be the trace character of the specified eigensheaf. For even k define

$$a_q(\lambda) = q + 1 - \#\mathcal{E}_\lambda(\mathbb{F}_q), \qquad M_{3,k}(q) = \sum_{\lambda \in \mathbb{F}_q \setminus \{0,1\}} \rho_q(\lambda) a_q(\lambda)^k.$$

The singular parameters are excluded throughout these moment sums.

Theorem 3.1. *For $r = 2, 4$, the representation H_r is continuous of dimension r over E, and is unramified outside the primes of K above $2, 3$. There is an exact sequence of $G_K = \mathrm{Gal}(\bar{K}/K)$-representations*

$$0 \longrightarrow E \longrightarrow H^1_c(U_{\bar{K}}, \mathcal{G}_r) \longrightarrow H^1(U_{\bar{K}}, \mathcal{G}_r) \longrightarrow E(-(r+1)) \longrightarrow 0. \tag{1}$$

Both middle spaces have dimension $r+1$, and their common parabolic image is H_r. At every $\mathfrak{p} \nmid 6$,

$$\mathrm{Tr}(\mathrm{Frob}_\mathfrak{p} \mid W_2) = -M_{3,2}(q) - q - 1, \tag{2}$$

$$\mathrm{Tr}(\mathrm{Frob}_\mathfrak{p} \mid V_4) = -M_{3,4}(q) + 3qM_{3,2}(q) + q^2 - 1. \tag{3}$$

These assertions include the degree-two prime ideals of K.

3.1 Local monodromy and the arithmetic boundary

At $\lambda = 0, 1$, the discriminant $16\lambda^2(1-\lambda)^2$ has valuation two, while $c_4 = 16(\lambda^2 - \lambda + 1)$ is a unit. Thus the fibres are multiplicative, with nontrivial unipotent inertia on $\mathcal{V}$. At infinity, the substitutions $t = 1/\lambda$, $x = \lambda X$, $y = \lambda^2 Y$ give

$$Y^2 = tX(X-1)(X-t).$$

This is the ramified quadratic twist of the multiplicative family at $t = 0$. The geometric inertia types on $\mathcal{V}$ at $0, 1, \infty$ are consequently $J_2(1), J_2(1), J_2(-1)$; see also [4, Chapter 15].

For clarity, the symmetric-power calculation is made over the characteristic-zero field E. In a local basis choose $N(e) = 0$, $N(h) = ce$, $c \neq 0$, for the unipotent logarithm. The induced logarithm satisfies

$$N_r(e^{r-j}h^j) = jc\, e^{r-j+1}h^{j-1}, \qquad N_r^r(h^r) = r!c^r e^r \neq 0.$$

It therefore has one block of length $r+1$. Since r is even, the scalar -1 at infinity disappears. No assertion that $r!$ is an integral unit is needed. With compatible tame generators the Kummer scalars at zero and infinity are a primitive cube root η and η^{-1}, respectively. Hence

point	$\mathcal{V}$	$\mathcal{G}_r$	$\dim_E \mathcal{G}_r^I$
0	$J_2(1)$	$\eta J_{r+1}(1)$	0
1	$J_2(1)$	$J_{r+1}(1)$	1
∞	$J_2(-1)$	$\eta^{-1} J_{r+1}(1)$	0

The same calculation holds in every good geometric fibre. The integral tameness argument below verifies that no Swan terms occur there.

Lemma 3.2. *The inertia invariant and coinvariant lines at $\lambda = 1$ are arithmetically*

$$\mathcal{G}_r^{I_1} \simeq E, \qquad (\mathcal{G}_r)_{I_1} \simeq E(-r).$$

In particular geometric Frobenius acts on the invariant line by 1, both over K and at every good residue field.

Proof. Put $v = x - 1$ in the singular fibre at $\lambda = 1$. Its equation $y^2 = (1+v)v^2$ has the two rational tangents $y = \pm v$, so its multiplicative reduction is split. The Tate torsion sequence, followed by inverse limit and duality, gives the cohomological sequence

$$0 \longrightarrow E \longrightarrow \mathcal{V} \longrightarrow E(-1) \longrightarrow 0$$

on the local punctured trait; see [5, pp. 18–19]. Its nontrivial unipotent inertia fixes the first line. In its r-th symmetric power, the invariant line is the r-th power of this trivial line, whereas the coinvariant quotient is the r-th power of the quotient $E(-1)$. Finally the torsor fibre over 1 is μ_3. It is rational over K and over each residue field at $\mathfrak{p} \nmid 6$, so its chosen character line is arithmetically trivial. Tensoring by that line gives both assertions. □

3.2 Cohomology and the two boundary terms

In this subsection cohomology is first taken over $\bar{K}$; the same arguments apply over good algebraically closed residue fields. The absence of I_0-invariants implies $H^0(U, \mathcal{G}_r) = 0$. It also implies $H^0(U, \mathcal{G}_r^\vee(1)) = 0$, since a constant Tate twist is trivial on geometric inertia. A lisse section on a connected nonproper smooth curve cannot have proper support unless it is zero. Poincare duality [6, Theorem 24.1] therefore gives

$$H_c^0(U, \mathcal{G}_r) = H_c^2(U, \mathcal{G}_r) = 0. \tag{4}$$

Comparison with the complex thrice-punctured sphere, which retracts onto a graph with one vertex and two loops, gives ordinary Euler characteristic $-(r+1)$ for a rank-$(r+1)$ local system. Duality gives the same compact-support Euler characteristic. The specialization below transports the dimensions to the good fibres. In particular $\dim_E H_c^1(U, \mathcal{G}_r) = r + 1$.

Let $j : U_{\bar{K}} \hookrightarrow \mathbb{P}^1_{\bar{K}}$, and let i_1 be the inclusion of the point 1. Lemma 3.2 and the local invariant table give an exact sequence of ordinary etale sheaves

$$0 \longrightarrow j_!\mathcal{G}_r \longrightarrow j_*\mathcal{G}_r \longrightarrow i_{1*}E \longrightarrow 0.$$

Its cohomology sequence and the vanishing of $H^0(U, \mathcal{G}_r)$ give

$$0 \longrightarrow E \longrightarrow H_c^1(U, \mathcal{G}_r) \longrightarrow H^1(\mathbb{P}^1, j_*\mathcal{G}_r) \longrightarrow 0. \tag{5}$$

The edge map for Leray injects $H^1(\mathbb{P}^1, j_*\mathcal{G}_r)$ into $H^1(U, \mathcal{G}_r)$. The composite of this injection with (5) is the natural compact-support map. It follows that

$$H^1(\mathbb{P}^1, j_*\mathcal{G}_r) = H_r, \qquad \dim_E H_r = r.$$

These identifications and exact sequences are Galois equivariant.

To determine the right boundary, the same sheaf sequence and (4) imply $H^2(\mathbb{P}^1, j_*\mathcal{G}_r) = 0$. Leray now gives a surjection from $H^1(U, \mathcal{G}_r)$ onto $\bigoplus_{s=0,1,\infty} H^1(I_s, \mathcal{G}_r)$, with kernel H_r. At $0, \infty$, the operator $T_s - 1$ is invertible, so the corresponding local groups vanish. At 1, tame inertia cohomology identifies the local group with

$$H^1(I_1, \mathcal{G}_r) \simeq (\mathcal{G}_r)_{I_1}(-1) \simeq E(-(r+1)).$$

The additional twist comes from the dual of the tame factor $\mathbb{Z}_2(1)$; geometric Frobenius acts on this factor of the local cohomology by q. Combining these observations proves (1), including its final surjectivity, and $\dim_E H^1(U, \mathcal{G}_r) = r + 1$.

3.3 Integral tame specialization

We give the specialization argument explicitly because U is not proper. The eigensheaf has an integral lattice $\mathcal{L}_{\rho,\mathcal{O}}$: the order-three projector is integral since 3 is invertible in $\mathcal{O}$. Put

$$\mathcal{G}_{r,\mathcal{O}} = \left(\mathrm{Sym}^r(R^1\pi_*\mathbb{Z}_2) \otimes_{\mathbb{Z}_2} \mathcal{O}\right) \otimes_{\mathcal{O}} \mathcal{L}_{\rho,\mathcal{O}}, \qquad \mathcal{G}_{r,m} = \mathcal{G}_{r,\mathcal{O}}/2^m.$$

The finite monodromy of $\mathcal{G}_{r,m}$ factors through the image of a subgroup of $\mathrm{GL}_2(\mathbb{Z}/2^m) \times \mu_3$, whose order is

$$18 \cdot 2^{4(m-1)}.$$

Thus for every residue characteristic $p \geq 5$, wild pro-p inertia at a boundary point acts trivially. In characteristic zero there is no wild inertia. This verifies relative tameness in the fibrewise discrete-valuation sense of [2, Expose XIII, Definition 2.1.1]. The sheaves are already lisse on U.

Apply [2, Expose XIII, Corollary 2.9] to $\mathbb{P}^1_S$, its relative normal-crossing boundary consisting of the three disjoint sections, and the additive coefficient sheaf $\mathcal{G}_{r,m}$. The compactification is smooth,

proper, and of finite presentation. The coefficient sheaf is finite, locally constant, constructible, relatively tame, and 2-primary, with 2 invertible on S. The theorem supplies local constancy and base change in degrees at most one. The same argument applies to the integral dual coefficient systems and their Tate twists; their additional finite cyclotomic factors have 2-power order.

On a geometric fibre, the finite H^0 groups form a Mittag–Leffler system. The continuous-cohomology comparison [6, Sections 18–21] consequently gives

$$H^1(U, \mathcal{G}_r) = \left(\varprojlim_m H^1(U, \mathcal{G}_{r,m}) \right) \otimes_{\mathcal{O}} E.$$

The finite-level specialization isomorphisms therefore pass to the E-adic groups. This argument does not require finite-level cohomology to be free.

For compact supports we use relative duality, rather than applying smooth proper base change to the open curve. The morphism $g : U \to S$ is smooth, separated, compactifiable, and of relative dimension one; S is quasi-compact and quasi-separated, and every 2^m is invertible on it. The relative trace and smooth-duality theorems [3, Expose XVIII, Theorem 2.9, Var. 2, and Theorem 3.2.5] apply. Their base-change compatibility, followed by passage to E, identifies

$$H^1_c(U, \mathcal{G}_r) \simeq H^1(U, \mathcal{G}_r^\vee(1))^\vee$$

compatibly with specialization. Moreover the natural transformation $Rg_! \to Rg_*$ is compatible with these base-change maps. Accordingly the specialization square with horizontal maps $H^1_c \to H^1$ commutes, and both vertical maps are isomorphisms. Taking images over E gives an isomorphism of the parabolic cohomology groups, equivariant for residue-field Galois action.

The compatible finite systems give a continuous representation factoring through $\pi_1(S)$. Their parabolic image is a stable subquotient, so H_r is unramified outside $2, 3$. At a good prime its Frobenius is identified with the Frobenius of the special-fibre parabolic group. This proves precisely the specialization needed for the arithmetic trace calculation.

3.4 The two trace formulas

Let α, β be the geometric Frobenius eigenvalues of a smooth Legendre fibre over $\mathbb{F}_q$. Thus $\alpha + \beta = a_q(\lambda)$ and $\alpha\beta = q$. The symmetric-power recurrence gives

$$\mathrm{Tr}(\mathrm{Frob}_\lambda \mid \mathrm{Sym}^2 \mathcal{V}) = a_q(\lambda)^2 - q, \qquad \mathrm{Tr}(\mathrm{Frob}_\lambda \mid \mathrm{Sym}^4 \mathcal{V}) = a_q(\lambda)^4 - 3q a_q(\lambda)^2 + q^2.$$

The compact-support trace formula [7], the vanishings (4), and the arithmetic boundary sequence (5) therefore give

$$\mathrm{Tr}(\mathrm{Frob}_q \mid H_r) = -\sum_{\lambda \neq 0,1} \mathrm{Tr}(\mathrm{Frob}_\lambda \mid \mathcal{G}_{r,\lambda}) - 1.$$

Since $\sum_{\lambda \neq 0,1} \rho_q(\lambda) = -1$, this is (2) for $r = 2$, and (3) for $r = 4$. The final -1 is the trace of the actual boundary line, not a correction inferred from its dimension. The preceding specialization identifies these special-fibre traces with the G_K-representations in Theorem 3.1, completing its proof.

For powers of a fixed Frobenius, the character on $\mathbb{F}_{q^d}^\times$ must be $\rho_q \circ N_{\mathbb{F}_{q^d}/\mathbb{F}_q}$. With this convention the same formulas compute $\mathrm{Tr}(\mathrm{Frob}_\mathfrak{p}^d \mid H_r)$. At an inert rational prime $p \equiv 2 \pmod 3$, the initial residue field is $\mathbb{F}_{p^2}$; its cubic character is not a nontrivial norm lift from $\mathbb{F}_p$. These formulas determine the trace over K and do not yet choose a trace at p for an extension to $G_\mathbb{Q}$.

Parameter inversion gives $a_q(1/\lambda) = \chi_q(\lambda) a_q(\lambda)$, because the two elliptic curves differ by the indicated quadratic twist. For even k, changing variables in the moment consequently yields

$$M_{3,k}(q; \rho_q) = M_{3,k}(q; \rho_q^{-1}) \in \mathbb{Z}.$$

Indeed the sum is a cyclotomic integer fixed by conjugation. This character independence does not itself supply coefficient descent.

Section 5 places both cubic eigenspaces in an ambient $\mathbb{Q}_2$-representation H_{nt}, constructs the arithmetic inversion operator I, and uses it to define the two extensions $W_4^{\pm}$ to $G_{\mathbb{Q}}$.

4 The cubic second moment and its modular comparison

Throughout this section the character has order three. Put

$$\mathcal{G}_2 = \mathrm{Sym}^2 \mathcal{V} \otimes \mathcal{L}_\rho, \qquad W_2 = H^1_!(U_{\overline{K}}, \mathcal{G}_2).$$

The preceding geometric construction gives $\dim_E W_2 = 2$, unramifiedness outside the primes above $2, 3$, and, for every other prime ideal $\mathfrak{p}$ with $q = N\mathfrak{p}$,

$$\mathrm{Tr}(\mathrm{Frob}_\mathfrak{p} \mid W_2) = -M_{3,2}(q) - q - 1. \tag{6}$$

Here Frobenius is geometric, $\tau = E(-1)$ has value q, and the Kummer normalization is the actual torsor $z^3 = \lambda$. We first obtain the remaining determinant constraint geometrically. The modular comparison then uses the separately recorded finite arithmetic certificate.

4.1 Arithmetic inversion and the determinant

Let $i(\lambda) = \lambda^{-1}$. On the etale cover $u^2 = \lambda$, the map

$$\phi_u : \mathcal{E}_\lambda \longrightarrow \mathcal{E}_{1/\lambda}, \qquad (x, y) \longmapsto (x/\lambda, y/u^3)$$

is an elliptic-curve isomorphism: substitution gives $(y/u^3)^2 = (x/\lambda)(x/\lambda - 1)(x/\lambda - 1/\lambda)$. Replacing u by $-u$ composes with $[-1]$, whose action on $\mathrm{Sym}^2 \mathcal{V}$ is trivial. Thus the induced isomorphism $i^* \mathrm{Sym}^2 \mathcal{V} \simeq \mathrm{Sym}^2 \mathcal{V}$ descends arithmetically. The map $z \mapsto z^{-1}$ on the cubic cover inverts the deck group, so $i^* \mathcal{L}_\rho \simeq \mathcal{L}_\rho^\vee$, E-linearly over K. This step does not conjugate the coefficient field.

The elliptic cup product induces the nondegenerate symmetric pairing

$$(v_1 v_2, w_1 w_2) \longmapsto \langle v_1, w_1 \rangle \langle v_2, w_2 \rangle + \langle v_1, w_2 \rangle \langle v_2, w_1 \rangle \quad \text{with values in } E(-2).$$

Consequently $i^* \mathcal{G}_2 \simeq \mathcal{G}_2^\vee(-2)$. Poincare duality [6, Theorem 24.1] induces a perfect pairing on parabolic images: the kernels of the maps $H^1_c \to H^1$ are the annihilators of the corresponding dual images. Hence

$$H^1_!(U, \mathcal{G}_2^\vee) \simeq W_2^\vee(-1), \qquad W_2 \simeq W_2^\vee(-3).$$

The inversion pullback commutes with G_K and with the compact-support map, so these are arithmetic isomorphisms. Taking determinants gives

$$\det W_2 = \tau^3 \varepsilon, \qquad \varepsilon^2 = 1, \tag{7}$$

where ε is continuous and unramified outside $2, 3$. In usual cyclotomic notation this is $\chi_{\mathrm{cyc}}^{-3} \varepsilon$. The assertion $\varepsilon = 1$ will use the finite comparison below.

4.2 The modular representation and residual conditions

Let $f = f_{36}$ be the weight-four normalized newform of level 36 and trivial nebentypus, labelled 36.4.*a*.*a*. Its eta expression is [12, Table 1]

$$f(z) = \frac{\eta(6z)^{14}}{\eta(2z)^3 \eta(18z)^3} - 3\eta(2z)^3 \eta(6z)^2 \eta(18z)^3. \tag{8}$$

Use the cohomological representation R_f over E: it is the dual of the conventional Deligne representation normalized at arithmetic Frobenius; see the modular-form motive and its etale realization in [22, Theorem 1.2.4(i)]. Thus geometric Frobenius at $p \geq 5$ on R_f has trace $a_f(p)$ and determinant p^3. Both W_2 and $R_f|_{G_K}$ are unramified outside primes above $2, 3$.

Their compact images admit stable $\mathcal{O}$-lattices, with $\mathcal{O}/(2) = \mathbb{F}_4$. Full rational 2-torsion on every smooth Legendre fibre over $\mathbb{F}_q$ makes $a_q(\lambda)$ even. Therefore $M_{3,2}(q) \in 4\mathcal{O}$, and (6)–(7) give

$$\operatorname{Tr} W_2 \equiv 0 \pmod 2, \qquad \det W_2 \equiv 1 \pmod 2.$$

Initially these congruences hold on good Frobenius classes; Chebotarev density and continuity extend them to G_K.

The carried certificate records, and the independent calculation described below rechecks, $f \equiv \eta(3z)^8 \pmod 2$ through the Sturm bound

$$\frac{4}{12}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(36)] = \frac{4}{12}\,72 = 24.$$

Here $\eta(3z)^8 \in S_4(\Gamma_0(9)) \subset S_4(\Gamma_0(36))$ by [12, Table 1]. Both sides have integral Fourier coefficients. Sturm's theorem [11, Theorem 1, p. 276] makes this a full Fourier-series congruence, subject to that recorded finite check. With $Q_z = e^{2\pi i z}$,

$$\eta(3z)^8 \equiv \eta(24z) = \sum_{m \geq 1} \chi_{12}(m) Q_z^{m^2} \pmod 2.$$

The congruence is the characteristic-two product identity; the equality is Euler's pentagonal-number identity, with $\chi_{12}(m) = (12/m)$ the Kronecker symbol. Prime-index coefficients of f are therefore even. At a split prime of K, the restricted modular trace is $a_f(p)$; at an inert prime it is $a_f(p)^2 - 2p^3$, again even. Its determinant is q^3. Density gives the required global congruences for $R_f|_{G_K}$ as well. The residue field $\mathbb{F}_4$ is allowed in the comparison criterion used below.

4.3 The carried finite certificate and its consequence

Let K_S be the compositum of the quadratic extensions of K unramified outside the primes above $2, 3$. The Eisenstein integer ring has class number one. Removing square factors from an allowed squareclass therefore leaves an S-unit. Its unit squareclass is generated by -1, and the primes above 2 and 3 can be generated by 2 and $\sqrt{-3}$. Their valuations, together with the nonsquare unit -1, prove independence. Consequently

$$K_S = K(\sqrt{-1}, \sqrt{2}, \sqrt{\sqrt{-3}}), \qquad \mathrm{Gal}(K_S/K) \simeq \mathbb{F}_2^3.$$

The seven nonzero vectors constitute a non-cubic set: a homogeneous cubic polynomial function vanishing on them also vanishes at zero, and hence vanishes on the whole vector space.

Table 1 reproduces the cubic rows of the carried exact certificate. A prime ideal is specified by p and the image r of $\sqrt{-3}$ modulo it. A bit is 0 for a square and 1 for a nonsquare; the displayed order is $(\sqrt{-3}, 2, -1)$, the reverse of the basis above.

The certificate consists of the supplied `V2_01_exact_livne.py`, its exact arithmetic helpers, `V2_exact_livne.txt`, and `exact_livne_certificate.json`. Static inspection shows direct finite-field character sums and integer eta-series arithmetic for the cubic comparisons, with no FFT rounding. The certificate uses the norm-lifted character over $\mathbb{F}_{p^2}$. Thus, by the geometric trace formula,

$$\begin{aligned} t &= -M_{3,2}(p) - p - 1, \\ u &= -M_{3,2}(p^2) - p^2 - 1 = \operatorname{Tr}(\mathrm{Frob}_{\mathfrak{p}}^2 \mid W_2), \\ d &= \frac{t^2 - u}{2}. \end{aligned}$$

p	r	bits	$t = a_f(p)$	d
7	2	001	8	343
7	5	101	8	343
13	6	110	−10	2197
19	4	011	−100	6859
19	15	111	−100	6859
37	16	010	−226	50653
73	17	100	26	389017

Table 1: Previously recorded exact cubic comparisons, with $d = p^3$. The seven rows and Sturm comparison have also been recomputed by a new standalone exact implementation.

The inversion identity for even Legendre moments makes the two primitive cubic-character choices equivalent here. The recorded values satisfy $t = a_f(p)$ and $d = p^3$ at every listed prime; the recorded Sturm comparison reaches degree 24. *The archived programs were not rerun.* A new standalone implementation, `independent_cubic_exact.py`, independently recomputes the displayed cubic comparisons using direct finite-field character sums and integer eta-series arithmetic. It uses only the Python standard library; the mathematical calculations use arbitrary-precision integers, without FFT or floating-point arithmetic. Over $\mathbb{F}_{p^2} = \mathbb{F}_p[s]/(s^2 - d)$ it uses $\chi_{p^2}(a + bs) = \chi_p(a^2 - db^2)$ and the norm-lifted cubic character. The new log records agreement at the five rational primes underlying the seven rows, all seven square-class vectors, and the congruence through degree 24. Historical expected values are read only after the new calculation, for comparison. Checks include every rational-fibre quadratic base-change identity and a second direct multiplication calculation over $\mathbb{F}_{49}$. Source hashes and the new log are supplied in the release supplement. This rechecks the bounded arithmetic; it does not re-prove the geometric hypotheses, newform identification, or external comparison theorems. The theorem below retains its explicit finite-check dependencies.

Theorem 4.1 (Cubic modular comparison using the carried certificate)**.** *Assuming the finite exact comparisons and the Sturm comparison just specified, the representations have isomorphic semisimplifications:*

$$W_2^{\mathrm{ss}} \simeq (R_f|_{G_K})^{\mathrm{ss}}, \qquad \det W_2 = \tau^3.$$

For every good prime ideal $\mathfrak{p}$ *of* K*,*

$$M_{3,2}(N\mathfrak{p}) = -\operatorname{Tr}(\mathrm{Frob}_{\mathfrak{p}} \mid R_f|_{G_K}) - N\mathfrak{p} - 1.$$

In particular, for $p \geq 5$ *and* $p \equiv 1 \pmod 3$*,* $M_{3,2}(p) = -a_f(p) - p - 1$*; for* $p \equiv 2 \pmod 3$*,*

$$M_{3,2}(p^2) = -a_f(p)^2 + 2p^3 - p^2 - 1.$$

Proof. Livne's criterion in [10, Theorem 4.1] applies to continuous rank-two representations over a finite extension of $\mathbb{Q}_2$. The common ramification set, vanishing residual traces, and equal residual determinants were established above. The seven rows give exact trace and determinant agreement on a non-cubic set in $\mathrm{Gal}(K_S/K)$. The criterion gives the asserted semisimplification. If it is formulated using the opposite Frobenius convention, invert both matrices: $\operatorname{Tr}(A^{-1}) = \operatorname{Tr}(A)/\det A$, and the quadratic test classes are unchanged under inversion.

Alternatively to taking the determinant of the comparison, (7) and the table give $\varepsilon(\mathrm{Frob}_{\mathfrak{p}}) = d/p^3 = 1$ on all seven nonzero classes. Since ε factors through $\mathrm{Gal}(K_S/K)$, it is trivial. Finally apply (6); restriction at an inert rational prime replaces Frobenius by its square. □

This section proves the stated cubic second-moment comparison with its explicit certificate dependency. It supplies no geometric proof for characters of orders four or six, and no bad-prime Euler-factor or rank-four modularity assertion.

5 Arithmetic involution, descent and polarization

We construct two four-dimensional representations over $\mathbb{Q}_2$ whose restrictions to G_K, after extending coefficients to E, are V_4. Throughout this section $\chi = \chi_{-3}$ is the quadratic character of $K/\mathbb{Q}$, and τ denotes the character of $\mathbb{Q}_2(-1)$.

5.1 The rational ambient representation

Over $\mathbb{Q}$, let

$$f : T = \mathbb{P}^1_z \setminus \{0, \infty, \mu_3\} \longrightarrow U, \qquad \lambda = z^3, \qquad \mathcal{A}_0 = \mathrm{Sym}^4(R^1\pi_*\mathbb{Q}_2).$$

The divisor μ_3 and the finite etale map f are defined over $\mathbb{Q}$. Set

$$H = H^1_!(T_{\overline{\mathbb{Q}}}, f^*\mathcal{A}_0), \qquad A = a^*, \quad a(z) = \zeta_K z, \qquad e_{\mathrm{nt}} = \frac{2 - A - A^2}{3}, \quad H_{\mathrm{nt}} = e_{\mathrm{nt}}H.$$

Here $\zeta_K \in K$ is a chosen primitive cube root, and A means pullback, with the identity identification $a^*f^*\mathcal{A}_0 = f^*\mathcal{A}_0$. The operator A is $\mathbb{Q}_2$-linear, although a is defined over K. If R denotes the Galois action on H, conjugation of the deck map gives

$$A^3 = 1, \qquad R(g)AR(g)^{-1} = A^{\chi(g)}. \tag{9}$$

Consequently e_{nt} and its image are $G_{\mathbb{Q}}$-stable.

Finite pushforward is exact, and the projection formula identifies $f_*f^*\mathcal{A}_0$ with $\mathcal{A}_0 \otimes f_*\mathbb{Q}_2$. These assertions can be checked on the finite sums of geometric stalks; see also [8, Proposition 59.55.2, tag 03QP]. Since f is proper, the same identification applies to compactly supported cohomology and commutes with $H^1_c \to H^1$. Thus it also identifies their parabolic images. With the chosen coefficient root $\zeta_E \in E$, take $\mathcal{L}_\rho$ to be the $A = \zeta_E$ eigensheaf of f_*E, as in the preceding construction. It follows that

$$H_{\mathrm{nt}} \otimes_{\mathbb{Q}_2} E = V_{4,\rho} \oplus V_{4,\rho^{-1}}, \qquad \dim_{\mathbb{Q}_2} H_{\mathrm{nt}} = 8. \tag{10}$$

The dimension follows from the previously proved dimension four of each character summand. Both summands are unramified at primes of K above $p \geq 5$. Since $K/\mathbb{Q}$ is unramified at such p, the inertia group of $\mathbb{Q}$ lies in G_K, so H_{nt} is also unramified there.

5.2 An actual coefficient involution

Lemma 5.1. *There is a $\mathbb{Q}_2$-linear operator I on H_{nt} such that*

$$I^2 = 1, \qquad IR(g) = R(g)I, \qquad IAI^{-1} = A^{-1}.$$

It is induced by an arithmetic coefficient isomorphism over inversion $z \mapsto z^{-1}$ and respects the compact-support and boundary maps.

Proof. Put $i(\lambda) = 1/\lambda$. On the finite etale double cover $u^2 = \lambda$ of U, the zero-preserving elliptic isomorphism

$$\phi_u : \mathcal{E}_\lambda \longrightarrow \mathcal{E}_{1/\lambda}, \qquad (x, y) \longmapsto (x/\lambda, y/u^3) \tag{11}$$

is verified by

$$(x/\lambda)(x/\lambda - 1)(x/\lambda - 1/\lambda) = x(x-1)(x-\lambda)/\lambda^3 = (y/u^3)^2.$$

Replacing u by $-u$ composes this map with $[-1]$, which acts as $+1$ on $\mathrm{Sym}^4 R^1\pi_*\mathbb{Q}_2$. The two pullbacks therefore agree on the double-cover relation and descend to an isomorphism

$$\theta : i^*\mathcal{A}_0 \longrightarrow \mathcal{A}_0$$

over $\mathbb{Q}$; in fact all these maps are defined over $\mathbb{Z}[1/6]$. For the second inversion choose $1/u$. Directly on both coordinates, $\phi_{1/u} \circ \phi_u = \mathrm{id}$, whence

$$\theta \circ i^*\theta = \mathrm{id}\,. \tag{12}$$

Choosing $-1/u$ gives the same identity on fourth symmetric powers.

Let $b(z) = 1/z$. Since $fb = if$, pullback of θ gives $\Theta : b^*f^*\mathcal{A}_0 \to f^*\mathcal{A}_0$. Define $I = \Theta_* b^*$ on cohomology. The map b extends to an automorphism of $\mathbb{P}^1_z$ preserving its boundary as a set. Its pullback and the coefficient isomorphism therefore act on H^1_c, H^1, and the natural map between them; they act on the parabolic image and on the boundary exact sequence as well. Equation (12) gives $I^2 = 1$. Both b and Θ are defined over $\mathbb{Q}$, so I commutes with R. Finally $bab = a^{-1}$, while the coefficient map is pulled back from the λ-line and the deck coefficient maps are identities. This proves $IAI^{-1} = A^{-1}$ on coefficient systems and hence on cohomology. It also proves that I preserves H_{nt} and exchanges the summands in (10). □

5.3 The two descents and the quadratic twist

Theorem 5.2. *The spaces*

$$W_4^+ = H_{\mathrm{nt}}^{I=1}, \qquad W_4^- = H_{\mathrm{nt}}^{I=-1}$$

are continuous four-dimensional $\mathbb{Q}_2$-representations of $G_{\mathbb{Q}}$, unramified outside $2, 3$, and

$$(W_4^{\pm}|_{G_K}) \otimes_{\mathbb{Q}_2} E \simeq V_4, \qquad W_4^- \simeq W_4^+ \otimes \chi.$$

Proof. First consider extension of the base field with coefficients E. Let $V = \ker(A - \zeta_E)$ in $H_{\mathrm{nt}} \otimes E$, and put $\delta(g) = (1 - \chi(g))/2$. For $g \notin G_K$, both $R(g)$ and I exchange the two eigenspaces. Hence

$$R_+(g) = I^{\delta(g)} R(g)|_V, \qquad R_-(g) = (-I)^{\delta(g)} R(g)|_V \tag{13}$$

preserve V. The commutation in Lemma 5.1 and additivity of δ modulo two show that these maps are representations. Their restriction to G_K is the original action.

Coefficient descent is supplied by the displayed rational eigenspaces. Indeed

$$\Phi_\pm : V \longrightarrow W_4^{\pm} \otimes E, \qquad v \longmapsto v \pm Iv$$

are isomorphisms whose inverses are projection to V. For $g \notin G_K$,

$$R(g)(v \pm Iv) = \Phi_\pm(\pm IR(g)v);$$

for $g \in G_K$ the intertwining is immediate. Thus these rational spaces realize exactly (13) and have dimension four. Continuity and the ramification assertion follow from H_{nt}.

For an explicit coefficient-descent datum, let $C = \mathrm{id} \otimes \sigma$, where $\sigma \in \mathrm{Gal}(E/\mathbb{Q}_2)$ sends ζ_E to ζ_E^{-1}. This semilinear operator commutes with A, I, R and exchanges the two eigenspaces. On V the operators $D_+ = IC$, $D_- = -IC$ are semilinear involutions commuting with the corresponding $R_\pm$. Under $\Phi_\pm$ they are ordinary coefficient conjugation. This checks the coefficient cocycle separately from base-field extension.

Finally put $J = A - A^2$ on H_{nt}. Since $A + A^2 = -1$ there,

$$J^2 = -3, \qquad IJ = -JI, \qquad R(g)JR(g)^{-1} = \chi(g)J.$$

Thus J is an invertible $\mathbb{Q}_2$-linear map from W_4^+ to W_4^-, intertwining the latter with $W_4^+ \otimes \chi$. □

Remark 5.3. The signs refer to the literal map (11): W_4^+ is its $+1$ eigenspace. Replacing I by $-I$ gives W_4^-. This constructs two particular extensions without any irreducibility assumption or classification of all extensions.

5.4 Boundary action

The boundary lines above $\lambda = 1$ have trivial arithmetic coefficient action. Formula (11) extends regularly there, since λ and u are units; at $u = 1$ it specializes to the identity. The choice $u = -1$ has the same effect on Sym^4. Consequently I acts on the three boundary lines by the permutation $z \mapsto z^{-1}$, with coefficient $+1$. The nontrivial deck boundary is therefore the augmentation subspace $B_{\mathrm{nt}} \subset \mathbb{Q}_2[\mu_3]$. The boundary contributions at $0, \infty$ are killed by e_{nt}, equivalently because the two twisted eigensheaves have no inertia invariants there. Writing $C_{\mathrm{nt}} = e_{\mathrm{nt}} H^1_c(T_{\overline{\mathbb{Q}}}, f^*\mathcal{A}_0)$, we obtain

$$0 \longrightarrow B_{\mathrm{nt}} \longrightarrow C_{\mathrm{nt}} \longrightarrow H_{\mathrm{nt}} \longrightarrow 0. \tag{14}$$

In B_{nt}, the lines generated by $2[1] - [\zeta_K] - [\zeta_K^2]$ and $[\zeta_K] - [\zeta_K^2]$ have I eigenvalues $+1, -1$ and Galois characters $1, \chi$, respectively. Thus, for $C_\pm = C_{\mathrm{nt}}^{I=\pm 1}$,

$$\begin{aligned} \mathrm{Tr}(\mathrm{Frob}_p \mid W_4^+) &= \mathrm{Tr}(\mathrm{Frob}_p \mid C_+) - 1, \\ \mathrm{Tr}(\mathrm{Frob}_p \mid W_4^-) &= \mathrm{Tr}(\mathrm{Frob}_p \mid C_-) - \chi(p) \end{aligned} \qquad (p \geq 5). \tag{15}$$

In particular the second boundary contribution is $+1$ at inert primes.

5.5 Alternating pairing and reciprocity

Proposition 5.4. *Each $W_4^\pm$ has a perfect alternating $G_{\mathbb{Q}}$-equivariant pairing with values in $\mathbb{Q}_2(-5)$. Its similitude character is τ^5 and its determinant is τ^{10}.*

Proof. The elliptic cup pairing $\langle\ ,\ \rangle$ on $R^1\pi_*\mathbb{Q}_2$ is alternating, perfect, and valued in $\mathbb{Q}_2(-1)$. On $\mathcal{A}_0$ put

$$B(v_1v_2v_3v_4, w_1w_2w_3w_4) = \sum_{\sigma \in S_4} \prod_{j=1}^{4} \langle v_j, w_{\sigma(j)} \rangle.$$

This is symmetric with values in $\mathbb{Q}_2(-4)$. In a symplectic basis e, f, its monomial matrix is antidiagonal with entries $(-1)^j j!(4-j)!\langle e, f\rangle^4$, so it is perfect over $\mathbb{Q}_2$. The degree-one elliptic isomorphisms ϕ_u preserve cup product; hence θ preserves B.

Poincare duality and the compact-support map give a perfect parabolic pairing $\Omega : H \times H \to \mathbb{Q}_2(-5)$ [6, Theorem 24.1]; see also the support and trace formalism of [3]. To see perfection on the image, the two maps $H^1_c \to H^1$ for dual coefficient systems are adjoint up to the degree-one cup sign, and their kernels are the annihilators of the opposite images. The degree-one cup sign and symmetry of B make Ω alternating. Both A and I preserve it: their coefficient maps are isometries and their curve automorphisms have degree one, so preserve the trace map on H^2_c.

The adjoint of A is A^{-1}; thus e_{nt} is self-adjoint and its image has nondegenerate restricted pairing. Since I is an involutive isometry, its two eigenspaces are orthogonal and each is nondegenerate. The restricted forms give the asserted equivariant pairings. Their multiplier is the character of $\mathbb{Q}_2(-5)$, namely τ^5. The square of a nondegenerate alternating two-form in dimension four is a volume form, yielding $\det W_4^\pm = \tau^{10}$. □

Corollary 5.5. *For $p \geq 5$, the characteristic polynomial of geometric Frobenius on either constructed representation is*

$$\det(X - \mathrm{Frob}_p \mid W_4^\pm) = X^4 - c_{1,\pm}(p)X^3 + c_{2,\pm}(p)X^2 - p^5 c_{1,\pm}(p)X + p^{10}.$$

The analogous formula for V_4 at a good prime of K uses its norm q in place of p.

Proof. The pairing identity $F^t \Omega F = p^5 \Omega$ pairs eigenvalues with their quotients p^5/α, and the determinant is p^{10}. These identities give the displayed coefficients. Restriction to G_K and scalar extension transfer the pairing to V_4. □

Remark 5.6. This argument establishes reciprocity without a purity premise. The coefficients of the displayed polynomials lie in $\mathbb{Q}_2$; a rational compatible-system assertion, Hodge numbers and bad-prime factors require their own arguments. Good-prime purity is proved next by a separate application of Deligne's weight theorem; the rational motivic realization and Hodge numbers are established in Section 6, and the local factors in Sections 8 and 9.

5.6 Good-prime Frobenius purity

Proposition 5.7 (Good-prime purity)**.** *For $r = 2, 4$, the representation H_r is pure of weight $r+1$ at every prime $\mathfrak{p} \nmid 6$ of K. More explicitly, if $q = N\mathfrak{p}$ and α is a geometric Frobenius eigenvalue on H_r, then α is algebraic over $\mathbb{Q}$ and $|\iota(\alpha)| = q^{(r+1)/2}$ for every complex embedding of its number field.*

Proof. Fix $\mathfrak{p} \nmid 6$, and work on $U_0 = \mathbb{P}^1_{\mathbb{F}_q} \setminus \{0, 1, \infty\}$. This is a smooth curve of finite type over $\mathbb{F}_q$, and the coefficient prime 2 is different from its characteristic. The specialized Legendre family is smooth and proper over U_0, so $\mathcal{V}_0 = R^1\pi_* E$ is lisse. At any closed point x of degree d, its geometric Frobenius eigenvalues are the two roots of $X^2 - a_{q^d}(x)X + q^d$; the Weil theorem for elliptic curves gives algebraic eigenvalues of absolute value $q^{d/2}$ under every complex embedding. Hence $\mathcal{V}_0$ is pointwise pure of weight one.

The cover $z^3 = \lambda$ is finite etale on U_0. Its chosen rank-one character summand is lisse and has finite monodromy, with Frobenius eigenvalues in μ_3 at every closed point; it is therefore pointwise pure of weight zero. Thus $\mathcal{G}_{r,0} = \mathrm{Sym}^r \mathcal{V}_0 \otimes \mathcal{L}_{\rho,0}$ is lisse and pure of weight r. The claims hold after extending coefficients to $\overline{\mathbb{Q}}_2$ and for every embedding used to define weights. Finite coefficient extension and passage to the specified direct summand preserve these algebraicity and absolute-value assertions.

Deligne's *Weil II*, Corollary 3.3.6 [21], states that for a smooth finite-type scheme and a lisse pure sheaf of weight w, the image of $H^i_c \to H^i$ is pure of weight $w + i$. Applying this result with $i = 1$ proves that the special-fibre parabolic group is pure of weight $r + 1$. Equivalently, on this curve one can apply Theorem 3.2.3 to $H^1(\mathbb{P}^1_{\overline{\mathbb{F}}_q}, j_*\mathcal{G}_{r,0})$. The specialization isomorphism of Section 3 is Frobenius equivariant and identifies this image with H_r, so the asserted arithmetic purity follows. Neither argument requires irreducibility or semisimplicity of H_r. □

Corollary 5.8. *Both W_4^+ and W_4^- are pure of weight five at every rational prime $p \geq 5$: each geometric Frobenius eigenvalue is algebraic and all its complex conjugates have absolute value $p^{5/2}$.*

Proof. By Theorem 5.2, the restrictions of $W_4^{\pm}$ to G_K, after extending coefficients, are H_4. At a split prime p the purity just proved applies to Frob_p itself. At an inert prime, Frob_p^2 restricts to the geometric Frobenius at the degree-two prime of K. Thus, if β is an eigenvalue of Frob_p on either $W_4^{\pm}$, then β^2 is algebraic and every complex conjugate has absolute value $(p^2)^{5/2} = p^5$. It follows that β is algebraic and all its complex conjugates have absolute value $p^{5/2}$, as required. □

Remark 5.9. This proposition concerns good-prime Frobenius weights of the specified 2-adic representations. It does not by itself identify their Hodge numbers, construct rational Scholl projectors or a compatible system, determine bad-prime Euler factors or conductors, or prove automorphy or analytic continuation. The separate construction in Section 6 supplies the specified rational Chow projectors and Hodge numbers. Sections 8–10 independently determine the bad-prime Euler factors and conductors.

5.7 Projector compatibility on the smooth locus

Lemma 5.10. *Let $\mathcal{F} = \mathcal{E} \times_U T$ and $h : Z = \mathcal{F}^{\times_T 4} \to T$. On relative cohomology set*

$$e_{\mathrm{odd}} = \prod_{j=1}^{4} \frac{1 - [-1]_j^*}{2}, \qquad e_{\mathrm{alt}} = \frac{1}{24} \sum_{\sigma \in S_4} \mathrm{sgn}(\sigma)\sigma^*, \qquad e_4 = e_{\mathrm{odd}} e_{\mathrm{alt}}.$$

These are rational commuting idempotents and

$$e_4 R^j h_* \mathbb{Q}_2 = 0 \quad (j \neq 4), \qquad e_4 R^4 h_* \mathbb{Q}_2 \simeq \mathrm{Sym}^4(R^1 \pi_* \mathbb{Q}_2)|_T.$$

The diagonal map induced by ϕ_u on $u^2 = \lambda$ commutes with e_4 and descends on this summand to the literal coefficient map θ of Lemma 5.1. On parabolic cohomology, $P_\pm = e_{\mathrm{nt}}(1 \pm I)/2$ are orthogonal rational idempotents with images $W_4^\pm$ and sum e_{nt}.

Proof. Elliptic inversion acts by -1 on R^1 and by $+1$ on R^0, R^2. Relative Kunneth therefore shows that e_{odd} selects exactly the fourfold tensor of R^1, in degree four. Permuting these four odd-degree factors introduces the Koszul sign. Consequently the signed permutation average becomes the ordinary symmetric tensor projector under Kunneth. This explains the sign used in Scholl's projector [22, Section 1.1.2].

The diagonal elliptic isomorphism commutes with individual inversions and factor permutations. Changing u to $-u$ composes it with simultaneous inversion on the four factors, acting by $(-1)^4 = 1$ on the selected summand. Its descended map is therefore exactly the fourth symmetric power used to define θ; the cocycle follows from $\phi_{1/u}\phi_u = 1$. The deck map commutes with e_4. Since $IAI^{-1} = A^{-1}$, I commutes with e_{nt}, and the last assertions follow by multiplying the displayed idempotents. □

This lemma establishes the smooth-family compatibility. We now extend it to rational Chow projectors on a specified compactification, including the auxiliary-level translations and the parabolic boundary comparison.

6 Rational Scholl realization and Hodge numbers

Write $\zeta = \zeta_K$. In this section π also denotes the pullback of the Legendre family to the displayed open base. For any prime ℓ, put

$$H_\ell = H^1_!(T_{\overline{\mathbb{Q}}}, \mathrm{Sym}^4 R^1\pi_*\mathbb{Q}_\ell), \qquad W_{4,\ell}^\pm = (e_{\mathrm{nt}} H_\ell)^{I=\pm 1},$$

using the same literal deck and inversion maps as in Section 5. At $\ell = 2$ these are the already constructed $W_4^\pm$, without a change of sign or a Tate twist.

Theorem 6.1 (Rational Scholl realization)**.** *There are a smooth projective, possibly disconnected fivefold $X/\mathbb{Q}$ and orthogonal idempotents $p_\pm \in \mathrm{CH}^5(X \times_{\mathbb{Q}} X) \otimes \mathbb{Q}$ such that for every prime ℓ*

$$p_\pm H^j_{\text{ét}}(X_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell) = 0 \quad (j \neq 5), \qquad p_\pm H^5_{\text{ét}}(X_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell) \simeq W_{4,\ell}^\pm.$$

The corresponding Betti realization is compatible with Betti–etale comparison and has $h^{5,0} = h^{0,5} = 2$, with all other Hodge numbers zero. Thus $(X, p_\pm, 0)$ are rational Chow motives realizing the normalized descents. The construction does not require irreducibility.

The following construction proves the theorem. Its use of Scholl's standard projector theorem is separated from the explicit cover, rational descent and sign computations for the present family.

6.1 A semistable auxiliary surface

Use the finite etale double cover

$$S = \mathbb{P}^1_s \setminus \{0, \infty, \mu_6\} \longrightarrow T, \qquad z = s^2, \qquad \lambda = s^6.$$

On S the elliptic family is $E_s : y^2 = x(x-1)(x-s^6)$. Its Weierstrass invariants are

$$\Delta = 16s^{12}(1-s^6)^2, \qquad c_4 = 16(1-s^6+s^{12}).$$

In characteristic zero, these give an I_{12} fibre at $s = 0$ and an I_2 fibre at each of the six roots of $s^6 = 1$. At infinity put $t = 1/s$, $X = t^6x$, $Y = t^9y$; the equation becomes $Y^2 = X(X-1)(X-t^6)$, giving another I_{12} fibre. The minimal regular model is therefore semistable, with no other singular fibres. Its nonconstant j-invariant makes the family strictly nonconstant.

Now take the *full frame torsor*

$$S' = \mathrm{Isom}_S((\mathbb{Z}/12\mathbb{Z})^2, E[12]) \longrightarrow S.$$

It includes all frames, not just one fixed Weil-pairing component and not frames modulo their sign. It is a finite etale torsor under $G = \mathrm{GL}_2(\mathbb{Z}/12\mathbb{Z})$, defined over $\mathbb{Q}$, although it can be disconnected. Let $\overline{S}'$ be its smooth projective normalization over $\mathbb{P}^1_s$, and let $\overline{E}' \to \overline{S}'$ be the minimal regular elliptic model. Write $\pi' : \overline{E}'|_{S'} \to S'$ for its smooth restriction, and ω' for the Hodge line bundle on $\overline{S}'$.

Why level 12 meets every component. Around a geometric I_m cusp, the local monodromy on a full level-12 frame is conjugate to

$$\begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \pmod{12}.$$

Its order is $e = 12/\gcd(12, m)$, the geometric ramification index of S'. After base change and minimal regularization the fibre is I_{me}. Here $m = 12$ or 2, so $me = 12$ in both cases. In the Tate description, the points represented by $q^{a/12}\zeta_{12}^b$ meet all twelve components of an I_{12} fibre. The tautological frame makes these points sections. Thus the constant group $V = (\mathbb{Z}/12\mathbb{Z})^2$ extends into the smooth locus and meets every geometric fibre component. Any splitting needed in this local description can be checked after a finite extension; the torsor itself is rational.

These facts verify the two geometric hypotheses of Scholl [23, Section 2.2]. On each geometric component one can first enlarge its number field of constants, apply that result, and then retain the entire finite disjoint union. No connected-component choice is needed for the rational construction.

Form $(\overline{E}')^4_{\overline{S}'}$ and apply Scholl's canonical resolution; call the result X. It has dimension five. The resolution blows up the intrinsic strata determined by how many elliptic coordinates are nodes [22, Section 3.0]. The nodal local models suffice [23, Section 2.10]. Consequently every family isomorphism considered below preserves its centers and lifts to X.

6.2 Literal automorphisms and the sign relation

On the family over S, define

$$\begin{aligned} a(s,x,y) &= (\zeta^2 s, x, y), & d(s,x,y) &= (-s, x, y), \\ b(s,x,y) &= (s^{-1}, x/s^6, y/s^9), & c(s,x,y) &= (s, x, -y). \end{aligned}$$

Use the same maps diagonally on four fibre coordinates. The order-three choice $a(s) = \zeta^2 s$ covers the required $z \mapsto \zeta z$. For b, the chosen square root is exactly $u = s^3$.

A point of the frame cover carries a frame $v : V \to E_s[12]$. Lift a and d by retaining this frame under the identity elliptic identification, and lift b by transporting it to $\phi_{s^3} \circ v$. For $h \in G$, let h

act by reparameterizing the frame while leaving elliptic points fixed. Write H_- for $h = -\mathrm{Id}$ and C for simultaneous inversion of all four fibre points, with the frame fixed. Upper-case A, B, D denote the pullback correspondences associated to the lower-case maps; g^* is represented by the transpose of its graph, with the pullback composition convention used throughout.

The coordinate formulas give the following identities, already on the open family with frames:

$$A^3 = 1, \qquad B^2 = D^2 = 1, \qquad AD = DA, \qquad BAB = A^{-1},$$
$$\boxed{BDB = H_-CD.}$$

For the last formula, applying b, d, b sends $(s, v, P_1, \dots, P_4)$ to $(-s, -v, -P_1, \dots, -P_4)$. Omitting the frame sign or asserting $BD = DB$ on the whole space would be an error. Both signs will disappear on the precise summand used below.

Extension across the cusps. An isomorphism of the generic elliptic curve extends to its minimal regular model by uniqueness [24]. This applies also to the translated sections and to base automorphisms permuting cusps. The local extensions agree on the dense open set and hence glue. The maps then preserve the nodal strata used in the canonical fibre-product resolution, so all act as genuine automorphisms of X, and their displayed relations remain exact there. This uses that specific resolution, not an arbitrary one.

Frame reparameterizations commute with A, B, D. They normalize the universal translation subgroup. The maps A, B, D transport the frame and fibre points together, so they normalize the translations, individual inversions, and permutations, preserving the character used in the following construction.

All relevant automorphisms generate a finite group: the translation, inversion, permutation and frame groups are finite, and the displayed relations give a finite quotient generated by A, B, D. We may therefore use their rational group algebra and its graph correspondences without a conjectural lifting of cohomological idempotents.

6.3 Exact rational Chow projectors

Let

$$\Gamma = (V \rtimes \{\pm 1\})^4 \rtimes \mathfrak{S}_4.$$

Its character ϵ is trivial on translations, is the product of the four inversion signs, and is sgn on $\mathfrak{S}_4$. Define

$$\Pi_\epsilon = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \epsilon(\gamma)\gamma^*, \qquad e_G = \frac{1}{|G|} \sum_{h \in G} h^*, \qquad q = e_G \Pi_\epsilon.$$

The signed permutation is essential: the Koszul sign from exchanging degree-one cohomology factors makes this select $\mathrm{Sym}^4 R^1$, rather than its alternating power. The translation average supplies the component condition at the boundary.

The group G normalizes Γ and preserves ϵ. Thus e_G and Π_ϵ commute, and $q^2 = q$. Moreover q commutes with A, B, D. There are exact identities in the correspondence algebra

$$H_- q = q, \qquad Cq = \epsilon(C)q = (-1)^4 q = q.$$

Hence $BDBq = Dq$, and $BDq = DBq$. Define

$$d_+ = \frac{1 + D}{2}, \qquad e_{\mathrm{nt}} = \frac{2 - A - A^2}{3}, \qquad b_\pm = \frac{1 \pm B}{2}$$

Their restrictions qd_+, qe_{nt}, and $qb_\pm$ are commuting idempotents in $q\mathbb{Q}[\mathcal{G}]q$, whose identity is q and where $\mathcal{G}$ is the finite group just constructed. Indeed B conjugates A to A^{-1}, so it commutes with e_{nt}; D commutes with A; and the remaining commutator has just been removed by q. Consequently

$$\boxed{p_\pm = q\,\frac{1 + D}{2}\,\frac{2 - A - A^2}{3}\,\frac{1 \pm B}{2}}$$

satisfy $p_\pm^2 = p_\pm$, $p_+p_- = p_-p_+ = 0$, and $p_+ + p_- = qd_+e_{\mathrm{nt}}$. These are equalities modulo rational equivalence, because they already hold for the finite sums of automorphism correspondences.

Descent to $\mathbb{Q}$. The frame torsor, X, B, D, and q are defined over $\mathbb{Q}$. Although A is defined over $\mathbb{Q}(\zeta)$, its Galois conjugate is A^2. Therefore $A + A^2$, e_{nt}, and $p_\pm$ are rational algebraic correspondences. More formally, restriction and corestriction of rational Chow cycles descend these invariant sums; restriction is injective after tensoring with $\mathbb{Q}$. Thus the idempotent equalities hold over $\mathbb{Q}$, not only over $\mathbb{Q}(\zeta)$.

No standard conjecture is used here. This is a decomposition by actual finite automorphisms; it does not assert a Hecke-eigenform Chow decomposition or any irreducibility of the resulting rank-four realizations by itself. Section 11 gives that separate argument for $\ell \neq 3$.

6.4 Parabolic realization and comparison

Scholl's finite-group construction [22, Theorem 1.2.1], in the semistable setting justified in [23, Sections 2.7–2.12], gives for Betti and etale cohomology

$$\Pi_\epsilon H^j(X) = 0 \quad (j \neq 5), \qquad \Pi_\epsilon H^5(X) \simeq H^1_!(S', \operatorname{Sym}^4 R^1\pi'_*\mathbb{Q}).$$

Here and below the coefficients on the etale side are $\mathbb{Q}_\ell$ and the base is geometrically extended. The identification is induced by restriction from the compact variety to the open fibre product, then relative Kunneth and Leray. It is compatible with every family automorphism above.

There is a useful descent subtlety. Some choices identifying the cusp terms in Scholl's boundary ladders become equivariant only after a finite extension. We do not use those chosen boundary identifications. The proper-to-open restriction, and the assertion that its image is the parabolic image, are natural. Their injectivity and image equality may be verified after a finite extension, and then hold for the same maps over $\mathbb{Q}$. The resulting identification is fully $G_{\mathbb{Q}}$-equivariant. Nor is it necessary, or correct, to say that every boundary cohomology group is killed by Π_ϵ.

For a finite etale torsor with a finite group, pullback and trace identify rational cohomology downstairs with invariants upstairs. The statement holds for both ordinary and compactly supported cohomology, commutes with the natural map between them, and therefore holds for its image. First take G invariants and then D invariants. This gives

$$qH^5(X) \simeq H^1_!(S, \operatorname{Sym}^4 R^1\pi_{S*}\mathbb{Q}),$$
$$qd_+H^5(X) \simeq H^1_!(T, \operatorname{Sym}^4 R^1\pi_*\mathbb{Q}).$$

On the latter space A is exactly the manuscript's cubic deck pullback. The map B uses $u = s^3$ in ϕ_u, so its induced map is exactly I. The discrepancy under $s \mapsto -s$ is simultaneous elliptic inversion, already killed on Sym^4. Applying e_{nt} and $b_\pm$ proves

$$p_\pm H^5(X) \simeq (e_{\mathrm{nt}} H^1_!(T, \operatorname{Sym}^4 R^1\pi_*\mathbb{Q}))^{I=\pm 1}.$$

This proves the theorem's etale assertion for every ℓ, in particular the required equality with the existing 2-adic $W_4^\pm$.

All maps used are pullback, trace, algebraic automorphisms, and natural cohomological maps. Betti–etale comparison is functorial for these operations, so the same formula identifies the rational Betti realization. Since it is a projected subspace of H^5 of a smooth projective variety, its Hodge structure is pure of weight five. Its precise Hodge numbers are computed next.

6.5 Hodge numbers and an independent sign check

The Legendre classifying map to the moduli *stack* of elliptic curves is etale: it factors as a C_2-torsor over the full-level-2 stack followed by an $\mathfrak{S}_3$-torsor [25, Lemma 4.3 and Proposition 4.5].

This includes the elliptic points where the coarse j-map ramifies. The maps $s \mapsto s^6$ and $S' \to S$ are etale on the open bases. Apply Scholl [23, Section 2.13] on S', where all the section hypotheses were verified, then take G invariants. The parabolic Hodge types are only $(5,0)$ and $(0,5)$, with

$$H^{5,0}_!(S, \operatorname{Sym}^4 R^1\pi_{S*}) \simeq H^0(\mathbb{P}^1_s, \omega^4 \otimes \Omega^1_{\mathbb{P}^1_s}),$$

where ω is the Hodge line bundle on $\mathbb{P}^1_s$. To justify this section formula after descent, write $r : \overline{S}' \to \mathbb{P}^1_s$. Semistable base change gives $\omega' = r^*\omega$, and $(r_*\Omega^1_{\overline{S}'})^G = \Omega^1_{\mathbb{P}^1_s}$; locally this is the identity $ds = eu^{e-1}du$ for $s = u^e$ in characteristic zero. The projection formula then gives the displayed equality of holomorphic spaces.

The singular-fibre multiplicities sum to $12+12+6\cdot 2 = 36$, giving $\deg \omega = 3$ and $\omega^4 \otimes \Omega^1 \simeq \mathcal{O}(10)$. The same degree follows directly from the two minimal charts. Write $\omega_s = dx/y$ and $\omega_t = dX/Y$ for relative invariant differentials. On their overlap $\omega_s = t^3\omega_t$. Thus

$$\beta_j = s^j\, ds\, \omega_s^4 = -t^{10-j}\, dt\, \omega_t^4, \qquad 0 \le j \le 10,$$

are a basis of holomorphic sections; the exponents show regularity at infinity. The fourth power denotes the symmetric coefficient power, not a fourth wedge of a one-form on the same elliptic curve.

Since $D^*\beta_j = (-1)^{j+1}\beta_j$, its invariant part has $j = 1, 3, 5, 7, 9$. With $z = s^2$ and $\omega_z = dx/y$ on the open family over T, let

$$\alpha_k = z^k\, dz\, \omega_z^4 \quad (0 \le k \le 4); \qquad (s \mapsto s^2)^*\alpha_k = 2\beta_{2k+1}.$$

These give the five-dimensional $(5,0)$ part on T. The literal maps yield

$$A\alpha_k = \zeta^{k+1}\alpha_k, \qquad I\alpha_k = -\alpha_{4-k}.$$

For the second identity, use the target coordinate s': $B^*\omega_{s'} = s^3\omega_s$ and $B^*(ds') = d(s^{-1}) = -s^{-2}ds$, hence $B^*\beta_j = -\beta_{10-j}$. This fixes the sign without any appeal to numerical traces.

The projector e_{nt} removes precisely α_2. On the remaining four holomorphic dimensions the two eigenspaces are

$$\begin{array}{l|ll} I = +1 & \alpha_0 - \alpha_4, & \alpha_1 - \alpha_3 \\ I = -1 & \alpha_0 + \alpha_4, & \alpha_1 + \alpha_3. \end{array}$$

Each has dimension two. Rationality of $p_\pm$ and complex conjugation give two antiholomorphic dimensions as well. Endpoint Hodge type excludes all other (p,q). Each realization therefore has rank four and the asserted Hodge numbers, independently confirming the prior rank calculation.

The construction and the Hodge calculation complete the proof of Theorem 6.1.

7 Inert Frobenius traces and good Euler factors

Retain the Legendre family $\mathcal{E}_\lambda$, its cubic cover $f : T \to U$, and the operators A and I from Sections 3 and 5. Thus A is pullback by $z \mapsto \zeta_3 z$, while I is induced by $z \mapsto z^{-1}$ and the fourth symmetric power of

$$\phi_u : \mathcal{E}_\lambda \longrightarrow \mathcal{E}_{1/\lambda}, \qquad (t,y) \longmapsto (t/\lambda, y/u^3), \qquad u^2 = \lambda.$$

In particular, $I^2 = 1$, $IAI^{-1} = A^{-1}$, and I commutes with Galois. Write H_{nt} for the eight-dimensional nontrivial deck summand and $W_4^\pm = H_{\mathrm{nt}}^{I=\pm 1}$. Frobenius on cohomology always means geometric Frobenius.

Fix a prime $p \geq 5$ with $p \equiv 2 \pmod 3$. Choose a nonsquare $d \in \mathbb{F}_p^\times$, a root $r \in \mathbb{F}_{p^2}$ of $r^2 = d$, and a primitive cubic character $\rho : \mathbb{F}_{p^2}^\times \to \mu_3$. For $x \in \mathbb{F}_p$ set

$$\nu_x = x + r, \qquad E'_{x,d} : \quad Y^2 = X(X^2 - 2xX + x^2 - d),$$
$$a'_p(x) = p + 1 - \#E'_{x,d}(\mathbb{F}_p), \qquad P_4(a, p) = a^4 - 3pa^2 + p^2,$$

and define

$$S_p = \sum_{x \in \mathbb{F}_p} \rho(x + r) P_4(a'_p(x), p). \tag{16}$$

Every fibre in this sum is smooth: its three roots $0, x + r, x - r$ are distinct. Character values can be regarded in $\mathbb{Q}(\zeta_3)$ and then embedded in the coefficient field $E = \mathbb{Q}_2(\zeta_3)$.

Theorem 7.1. *The sum S_p is an integer independent of d, the sign of r, and the primitive cubic character. At every such prime,*

$$\mathrm{Tr}(\mathrm{Frob}_p \mid W_4^+) = -S_p - 1, \qquad \mathrm{Tr}(\mathrm{Frob}_p \mid W_4^-) = S_p + 1. \tag{17}$$

Proof. We realize the operator $I \mathrm{Frob}_p$ by an actual variety and sheaf over $\mathbb{F}_p$, so that the ordinary trace formula applies. Let

$$C = \ker\Big(N_{\mathbb{F}_{p^2}/\mathbb{F}_p} : \mathrm{Res}_{\mathbb{F}_{p^2}/\mathbb{F}_p} \mathbb{G}_m \longrightarrow \mathbb{G}_m\Big).$$

This torus has equation $a^2 - db^2 = 1$. Over $\mathbb{F}_{p^2}$, the coordinate $z = a + rb$ identifies C with $\mathbb{G}_m$. Since $r^p = -r$, arithmetic Frobenius on geometric points is

$$z \longmapsto z^{-p}. \tag{18}$$

Indeed $(a - rb)^p = a^p + rb^p$. Put $U' = C \setminus \{1\}$ and $T' = C \setminus C[3]$. The cube map $h : T' \to U'$ is finite étale of degree three. Its deck group $C[3]$ is constant over $\mathbb{F}_p$, because $\zeta^{-p} = \zeta$ for $\zeta \in \mu_3$. Over $\mathbb{F}_{p^2}$ the map is the original cover $z^3 = \lambda$. The parameter

$$\lambda = \frac{x + r}{x - r}, \qquad x = r\frac{\lambda + 1}{\lambda - 1} \tag{19}$$

identifies U' with $\mathbb{A}^1_{\mathbb{F}_p} \setminus V(x^2 - d)$. Thus $U'(\mathbb{F}_p) = \mathbb{F}_p$ in this parameter. The removed point $\lambda = 1$ corresponds to $x = \infty$; the other geometric punctures are $x = r, -r$.

Let $\mathcal{K}_{\mathrm{nt}}$ be the rank-two augmentation summand of $h_*\mathbb{Q}_2$. The elliptic curves $E'_{x,d}$ form an elliptic scheme $\pi' : \mathcal{E}' \to U'$, and we use the lisse sheaf

$$\mathcal{B} = \mathrm{Sym}^4(R^1\pi'_*\mathbb{Q}_2) \otimes \mathcal{K}_{\mathrm{nt}}. \tag{20}$$

We now check the coefficient descent on the whole family. Over $\mathbb{F}_{p^2}$ write $a = x - r$, $\nu = x + r$, so that $\lambda = \nu/a$ and $\bar{a} = \nu$. On the finite étale cover $s^2 = a$ there is an elliptic isomorphism

$$\alpha : \mathcal{E}' \longrightarrow \mathcal{E}_\lambda, \qquad (X, Y) \longmapsto (X/a, Y/s^3).$$

Replacing s by $-s$ composes this map with $[-1]$; hence its fourth symmetric power descends to $U'_{\mathbb{F}_{p^2}}$. On a common square-root cover choose $\bar{s}$ with $\bar{s}^2 = \bar{a} = \nu$ and put $u = \bar{s}/s$. Then $u^2 = \lambda$ and the maps of families satisfy

$$\phi_u \circ \alpha(X, Y) = (X/\nu, Y/\bar{s}^3) = \bar{\alpha}(X, Y). \tag{21}$$

This is exactly the compatibility of the induced pullbacks with the coefficient map defining I. All square-root choices disappear on Sym^4. Effective étale descent, applied at finite coefficient levels and then to their compatible inverse system, identifies (20) with the inversion twist of the original

coefficient system [9]. Its geometric Frobenius on cohomology is IF, where $F = \mathrm{Frob}_p$ on the original cohomology: the arithmetic descent uses the involution I, and $I^{-1} = I$ and $IF = FI$. In particular, at a rational point the elliptic coefficient trace is exactly $P_4(a'_p(x), p)$. This follows from the eigenvalue sum $\alpha^4 + \alpha^3\beta + \alpha^2\beta^2 + \alpha\beta^3 + \beta^4$ with $\alpha + \beta = a'_p(x)$ and $\alpha\beta = p$; no additional scalar occurs.

For the Kummer coefficient, fix $x \in \mathbb{F}_p$, choose $v^3 = \nu_x$, and take the lift $z = v^{1-p}$ of λ. Then $z^3 = \nu_x^{1-p} = \nu_x/\bar{\nu}_x$. The arithmetic point map (18) takes z to κz, where

$$\kappa = z^{-p-1} = v^{p^2-1} = \nu_x^{(p^2-1)/3} \in \mu_3. \tag{22}$$

The result is independent of the cube root v. For the pullback eigenspace of functions specified by $g(\zeta_3 z) = \zeta_E g(z)$, geometric Frobenius acts by pullback under this arithmetic point permutation and has eigenvalue $\iota(\kappa)$, where $\iota(\zeta_3) = \zeta_E$. Thus we may take $\rho(\nu) = \iota(\nu^{(p^2-1)/3})$; the other nontrivial line has inverse eigenvalue. The local trace of $\mathcal{B}$ is therefore

$$\big(\rho(x+r) + \rho(x+r)^{-1}\big) P_4(a'_p(x), p). \tag{23}$$

This computation distinguishes arithmetic action on points from geometric action on cohomology.

Put $\mathcal{A}_0 = \mathrm{Sym}^4 R^1\pi_*\mathbb{Q}_2$ and $C_{\mathrm{nt}} = e_{\mathrm{nt}} H^1_c(T_{\overline{\mathbb{F}}_p}, f^*\mathcal{A}_0)$, where $e_{\mathrm{nt}} = (2 - A - A^2)/3$. Finite étale pushforward identifies this with the corresponding compactly supported cohomology on the base [8]. The geometric groups in degrees zero and two vanish by Section 3, and twisting preserves those vanishings. The trace formula [7] and (23) give

$$\mathrm{Tr}(IF \mid C_{\mathrm{nt}}) = -(S_p + \bar{S}_p).$$

The boundary calculation of Section 3 gives

$$0 \longrightarrow B_{\mathrm{nt}} \longrightarrow C_{\mathrm{nt}} \longrightarrow H_{\mathrm{nt}} \longrightarrow 0, \qquad B_{\mathrm{nt}} = \mathbb{Q}_2[\mu_3]_{\mathrm{nt}}.$$

The coefficient line at $\lambda = 1$ has Frobenius eigenvalue 1, and the coefficient action of I there is $+1$. Both F and I permute μ_3 by inversion. Hence IF is the identity on this two-dimensional boundary, and

$$\mathrm{Tr}(IF \mid H_{\mathrm{nt}}) = -(S_p + \bar{S}_p) - 2. \tag{24}$$

Moreover, $FAF^{-1} = A^{-1}$, so F exchanges the two nontrivial A eigenspaces and $\mathrm{Tr}(F \mid H_{\mathrm{nt}}) = 0$. The projectors $(1 \pm I)/2$ now give

$$\mathrm{Tr}(F \mid W_4^+) = -\frac{S_p + \bar{S}_p}{2} - 1, \qquad \mathrm{Tr}(F \mid W_4^-) = \frac{S_p + \bar{S}_p}{2} + 1.$$

It remains to establish integrality and independence of choices. The character ρ is trivial on $\mathbb{F}_p^\times$, because $3 \nmid p - 1$. Under $x \mapsto -x$ we have $\nu_{-x} = -\bar{\nu}_x$ and hence $\rho(\nu_{-x}) = \rho(\nu_x)^{-1}$. The curve $E'_{-x,d}$ is the quadratic twist by -1 of $E'_{x,d}$; therefore $a'_p(-x) = \chi_p(-1)a'_p(x)$, where χ_p is the quadratic character. Since P_4 is even, $S_p = \bar{S}_p$. As $S_p \in \mathbb{Z}[\zeta_3]$, it follows that $S_p \in \mathbb{Z}$ and the trace identities follow. If $d' = c^2 d$ and $r' = cr$, relabel $x' = cx$. The character weight is unchanged, and $E'_{cx,d'}$ is the quadratic twist by c of $E'_{x,d}$, again invisible to P_4. Every two nonsquares differ by such a square. Changing the sign of r or replacing ρ by its inverse conjugates the sum and therefore also leaves it unchanged. □

For the good Euler factors, let

$$v(q) = -M_{3,4}(q) + 3qM_{3,2}(q) + q^2 - 1$$

be the V_4 trace established in Section 3. Whenever q is a higher power of the residue field of a fixed prime, the cubic character in these moments is its norm lift.

Corollary 7.2. *For every rational prime $p \geq 5$, put*

$$c_1(p) = \begin{cases} v(p), & p \equiv 1 \pmod 3, \\ S_p + 1, & p \equiv 2 \pmod 3, \end{cases} \qquad c_2(p) = \frac{c_1(p)^2 - v(p^2)}{2}.$$

Then the good Euler polynomial of W_4^- is

$$\det(1 - \mathrm{Frob}_p T \mid W_4^-) = 1 - c_1(p)T + c_2(p)T^2 - p^5 c_1(p)T^3 + p^{10}T^4. \tag{25}$$

Proof. At split primes, either extension restricts to V_4 and has trace $v(p)$. At inert primes Theorem 7.1 gives the stated first trace. In both cases $\mathrm{Tr}(\mathrm{Frob}_p^2 \mid W_4^-) = v(p^2)$: at a split prime this uses the norm-lifted character from the chosen residue field $\mathbb{F}_p$; at an inert prime $\mathbb{F}_{p^2}$ is the initial residue field. Newton's identity gives $c_2(p)$. The perfect alternating pairing of Section 5, with Frobenius multiplier p^5, gives determinant p^{10} and the coefficient $-p^5 c_1(p)$ of T^3. □

Thus the exact finite-sum prescription with inert first trace $S_p + 1$ is the good-prime Euler prescription of W_4^-. This statement does not verify numerical implementations or floating-point tables. The separate local proofs in Sections 8 and 9 supply the factors at 2 and 3. An identification of completed analytic L-functions remains outside the established scope.

8 Local inertia and the Euler factors at 3

This section computes the arithmetic local representation of the literal rational descents fixed in Section 5. It combines an explicit quotient with the local two-section vanishing-cycle calculation and a separate Frobenius computation. The proof does not use a numerical functional-equation fit.

8.1 Statement and normalization

Let $\ell \neq 3$, let

$$T = \mathbb{P}^1_z \setminus \{0, \infty, \mu_3\}, \qquad E_z : y^2 = x(x-1)(x-z^3), \qquad H_\ell = H^1_!(T_{\overline{\mathbb{Q}}}, \mathrm{Sym}^4 R^1\pi_* \mathbb{Q}_\ell).$$

Here $H^1_! = \mathrm{im}(H^1_c \to H^1)$, A is pullback by $z \mapsto \zeta_3 z$, and $e_{\mathrm{nt}} = (2 - A - A^2)/3$. The literal inversion I uses

$$z \mapsto z^{-1}, \qquad (x, y) \mapsto (x/z^3, y/u^3), \quad u^2 = z^3.$$

The square-root sign disappears on the fourth symmetric power. Put $W^{\pm}_{4,\ell} = (e_{\mathrm{nt}} H_\ell)^{I=\pm 1}$, as in [26]. We use geometric Frobenius, so $\mathbb{Q}_\ell(-1)$ has eigenvalue 3 at 3. The standing descent identities are

$$\dim W^{\pm}_{4,\ell} = 4, \qquad W^-_{4,\ell} \simeq W^+_{4,\ell} \otimes \chi_{-3}.$$

They follow from the same finite-cover and involution construction for every ℓ, without requiring compatibility at a bad prime.

Theorem 8.1. *For every $\ell \neq 3$, the local quantities of these specified representations are*

	$\dim(W^{\pm}_{4,\ell})^{I_3}$	$P_3^{\pm}(T)$	$\mathrm{Sw}_3(W^{\pm}_{4,\ell})$	$a_3(W^{\pm}_{4,\ell})$
+	1	$1 - 3T$	0	3
−	0	1	0	4

where $P_3^{\pm}(T) = \det(1 - \mathrm{Frob}_3 T \mid (W^{\pm}_{4,\ell})^{I_3})$. On $W^+_{4,\ell}$, inertia is unipotent with a single Jordan block of length four; on $W^-_{4,\ell}$ it is its ramified quadratic twist. If N is the nonzero tame monodromy logarithm and F is geometric Frobenius, then $FNF^{-1} = 3^{-1}N$. On the successive one-dimensional quotients of $0 \subset \ker N \subset \ker N^2 \subset \ker N^3 \subset V$ for $V = W^+_{4,\ell}$, the Frobenius eigenvalues are $3, 9, 27, 81$.

In particular, the original 2-adic representation can be used directly at 3. The argument also works at $\ell = 5$ and yields the same displayed data. The dyadic calculation is treated separately below.

8.2 An exact quotient, including the inversion sign

Set $t = z + z^{-1}$ and consider

$$C_t : Y^2 = X^3 + (t-1)(t+2)X^2 + (t+2)X. \tag{26}$$

Write $a = (t-1)(t+2)$, $b = t+2$. Its invariants are

$$\begin{aligned} c_4 &= 16(a^2 - 3b) = 16(t+2)(t^3 - 3t - 1), \\ \Delta &= 16b^2(a^2 - 4b) = 16(t+2)^3(t-2)(t+1)^2, \\ j &= \frac{256(t^3 - 3t - 1)^3}{(t-2)(t+1)^2}. \end{aligned}$$

Over $\mathbb{Q}(z)$ put $d = -z^2/(z+1)$. Direct identities give

$$da = -(1 + z^3), \qquad d^2 b = z^3.$$

Thus E_z is precisely the quadratic twist of C_t with coefficients da, d^2b. Over $v^2 = d$, the map

$$\psi_z : E_z \longrightarrow C_t, \qquad (x, y) \longmapsto (x/d, y/v^3)$$

induces a canonical fourth-symmetric-power isomorphism. Its two choices differ by $[-1]$, which acts trivially on that symmetric power.

Crucially, $d(1/z) = d(z)/z^3$. Choosing $v' = v/u$ for the inverted parameter gives

$$\psi_{1/z} \circ \phi_u(x, y) = (x/d, y/v^3) = \psi_z(x, y).$$

This proves compatibility with the *literal I* in the manuscript. The descent is the plus descent; it has no additional constant quadratic twist.

Let $V = \mathbb{P}^1_t \setminus \{2, -1, -2, \infty\}$ and $Z = \mathbb{P}^1_z \setminus \{0, \infty, \mu_3, -1\}$. Then $Z \to V$ is a finite etale double cover, and the displayed isomorphism identifies its invariant coefficient system with $\mathcal{F} = \mathrm{Sym}^4 R^1(C/V)_*\mathbb{Q}_\ell$. On the compact curves, finite pushforward followed by invariants identifies the ordinary middle-extension sheaves. This can be checked stalkwise: the stalk at a missing point is the invariant space of its punctured-trait monodromy. Finite pushforward and rational finite-group invariants are exact. Consequently

$$H^1(\mathbb{P}^1_{\bar{\mathbb{Q}}}, j_*\mathcal{F}) \simeq H_\ell^{I=+1}. \tag{27}$$

Adding back $z = -1$ on the original side makes no change to its parabolic cohomology, since its original coefficient sheaf is smooth there.

8.3 The rank and the extra branch point

The singular fibres of (26) in characteristic zero are

t	2	-1	-2	∞
type	I_1	I_2	III	I_6
$\dim \mathcal{F}^{I_t}$	1	1	3	1

At infinity use $r = 1/t$, $X' = r^2X$, $Y' = r^3Y$. The resulting coefficients are $1 + r - 2r^2$ and $r^3(1 + 2r)$; c_4 is a unit and Δ has order six. At -2, the orders of c_4, Δ are $1, 3$, giving type III.

Equivalently, the ramified preimage $z = -1$ has elliptic automorphism $(x, y) \mapsto (-x, iy)$. Its H^1 eigenvalues are $i, -i$, so the eigenvalues of its fourth symmetric power are $1, -1, 1, -1, 1$. This explains both the three-dimensional stalk at -2 and why that point must be included in the rank calculation.

The Legendre geometric monodromy is SL_2; finite-index pullback preserves its connected Zariski closure. The quotient identification therefore shows that neither $\mathcal{F}$ nor its dual has a geometric global invariant. The parabolic Euler-characteristic formula gives

$$\dim H^1(\mathbb{P}^1, j_*\mathcal{F}) = (4-2)5 - (1+1+3+1) = 4.$$

The already defined rank-four $W^+_{4,\ell}$ is a subspace of the right side of (27); equality of dimensions proves

$$W^+_{4,\ell} = H^1_!(V_{\bar{\mathbb{Q}}}, \mathcal{F}) \tag{28}$$

with its full arithmetic action. This step does not use Hodge numbers or a numerical Frobenius comparison.

8.4 The collision at 3 and its local cycles

On $\mathcal{X} = \mathbb{P}^1_{\mathbb{Z}_3}$ remove the four sections $D = \{2, -1, -2, \infty\}$ and let $j : \mathcal{U} \hookrightarrow \mathcal{X}$. The elliptic scheme (26) is smooth and proper over $\mathcal{U}$, since its discriminant is a unit there. Thus $\mathcal{F}$ is lisse on this entire open scheme for every $\ell \neq 3$.

The sections $t = 2, -1$ meet at the special point x with $t = -1 \in \mathbb{F}_3$. Their intersection multiplicity is $v_3(2 - (-1)) = 1$. The other two sections specialize to $1, \infty$, separately. Near x the invariant c_4 is a unit and the discriminant is a unit times $(t-2)(t+1)^2$. At x the nodal cubic is

$$Y^2 = X^3 + X^2 + X = X(X-1)^2.$$

The node $(1, 0)$ has rational tangents $Y = \pm(X - 1)$: the degeneration is split multiplicative. The two horizontal monodromies on R^1 have multiplicities one and two in the same Tate unipotent direction.

We apply *the local* theorem [23, §§1.6–1.9], not its global semistable-family Theorem 2.15. The present family has an additive point at -2, so that global theorem's hypotheses should not be asserted. For the local theorem, the base curve is smooth, the two sections have intersection multiplicity one, the coefficient system is lisse off them, and Tate uniformization makes its local monodromy factor through the two Kummer characters with multiplicities $1, 2$. These are exactly the required local hypotheses.

Put $B = (R^1\Phi j_!\mathcal{F})_x$. Scholl's calculation identifies the inertia action on B with the rank-five fourth symmetric power of a nontrivial unipotent rank-two action. Its logarithm is proportional to $(1+2)N_0 = 3N_0$, and B is a single unipotent block of length five. In particular wild inertia acts trivially. The local boundary sequence [23, §1.7] gives

$$0 \longrightarrow A_x \longrightarrow B \longrightarrow C_x \longrightarrow 0, \qquad \dim A_x = 1, \quad \dim C_x = 4. \tag{29}$$

Here A_x is the cokernel of the diagonal map from the common invariant line to the two cusp invariant lines. It is the unique invariant line of B. Thus C_x is a single unipotent block of length four.

8.5 The global-to-local surjection

This paragraph supplies the necessary bridge from local cycles to the specified global representation. It also avoids any assumption about the absence of other vanishing cycles at the additive point. In particular, we do not invoke the original deduction of Corollary 2.16 in Scholl's paper; its published correction is included in the reference [23].

On the special open curve $\mathcal{U}_{\bar{\mathbb{F}}_3}$, $H^2_c(\mathcal{U}_{\bar{\mathbb{F}}_3}, \mathcal{F}) = 0$. Indeed pull back by the separable double cover $t = z + z^{-1}$. The exact quadratic-twist identities still hold in characteristic three. After taking Sym^4, the resulting system is the Legendre system at z^3. The relative Frobenius isogeny from the Legendre curve at z to that at z^3 induces an isomorphism on ℓ-adic H^1 for $\ell \neq 3$. The Legendre geometric monodromy is SL_2 in every odd characteristic [4, Chapter 15, opening discussion]; hence its fourth symmetric power and its dual have no global invariants. Pullback detects invariant sections, and Poincare duality gives the stated vanishing.

Apply the nearby/vanishing-cycle triangle to $\mathcal{G} = j_!\mathcal{F}$ on the proper curve $\mathcal{X}$. Since $\mathcal{G}$ is lisse on the smooth open, $R\Phi\mathcal{G}$ is supported on the finite special boundary. Its degree-one hypercohomology is therefore the direct sum of its degree-one stalks. The exact sequence contains

$$H^1_c(V_{\bar{\mathbb{Q}}_3}, \mathcal{F}) \longrightarrow \bigoplus_{y \in D_{\bar{\mathbb{F}}_3}} (R^1\Phi\mathcal{G})_y \longrightarrow H^2_c(\mathcal{U}_{\bar{\mathbb{F}}_3}, \mathcal{F}) = 0.$$

Projection gives a surjection onto B. Naturality of the boundary maps and (29) show that the boundary subspace of H^1_c maps into A_x. More explicitly, the two distant cusp stalks have zero contribution at x, while the two colliding invariant lines map through $(\mathbb{Q}_\ell \oplus \mathbb{Q}_\ell)/\operatorname{diag}\mathbb{Q}_\ell = A_x$ and onto it. Thus there is an induced surjection

$$H^1_!(V_{\bar{\mathbb{Q}}_3}, \mathcal{F}) \longrightarrow C_x.$$

Its source and target have dimension four, so it is an isomorphism. All constructions are natural over $\mathbb{Q}_3$, and x is rational over $\mathbb{F}_3$; the isomorphism is equivariant for the full local Galois group. Together with (28), this proves that $W^+_{4,\ell}$ has tame, unipotent inertia with a single length-four block.

8.6 Arithmetic Frobenius: fixing the sign and the Tate twist

Scholl's local theorem by itself specifies inertia, not the arithmetic Frobenius eigenvalue. We give the extra calculation here. Near the split Tate degeneration, R^1 has its natural filtration

$$0 \longrightarrow \mathbb{Q}_\ell \longrightarrow R^1 \longrightarrow \mathbb{Q}_\ell(-1) \longrightarrow 0.$$

The fourth symmetric power has five graded pieces $\mathbb{Q}_\ell(-j)$, $0 \le j \le 4$. They are unramified arithmetic coefficient lines extending over the local base. The split-node tangent calculation removes any torus splitting character. Alternatively a quadratic splitting character disappears on the even fourth power.

For a constant unramified coefficient line L, the local vanishing cycles of extension by zero at two colliding, individually rational sections satisfy

$$R^1\Phi(j_!L)_x = \operatorname{coker}(L \xrightarrow{\mathrm{diag}} L \oplus L) \simeq L, \qquad R^q\Phi(j_!L)_x = 0 \quad (q \neq 1).$$

This follows naturally from the constant-sheaf/closed-boundary exact sequence and smooth base change. Because both sections are separately rational, Frobenius does not exchange them. In particular no sign character and no extra Tate twist enter this formula.

Apply the vanishing-cycle functor to the Tate filtration. The displayed concentration in degree one makes successive sequences short exact at this stalk. It yields an arithmetic filtration of B with graded pieces

$$\mathbb{Q}_\ell,\ \mathbb{Q}_\ell(-1),\ \mathbb{Q}_\ell(-2),\ \mathbb{Q}_\ell(-3),\ \mathbb{Q}_\ell(-4).$$

The boundary line A_x is the first piece: it is the difference of the two split cusp invariant lines. Hence C_x has graded pieces with $j = 1, 2, 3, 4$. Its first one-dimensional filtered piece is inertia invariant; the single length-four block has just one invariant line. Therefore

$$(W^+_{4,\ell})^{I_3} \simeq \mathbb{Q}_\ell(-1), \qquad P^+_3(T) = 1 - 3T.$$

This determines the positive sign arithmetically. It does not infer that sign from purity, polarization or a numerical functional equation.

The same filtration determines the full ordered monodromy flag. Every filtered piece is inertia-stable, and a single nilpotent Jordan block has a unique invariant subspace of each dimension. Thus these pieces are exactly $\ker N^r$, with graded Frobenius eigenvalues 3^r for $r = 1, 2, 3, 4$. Geometric Frobenius acts on the tame inertia parameter by 3^{-1}, so taking logarithms gives $FNF^{-1} = 3^{-1}N$.

8.7 The minus descent and the conductor exponents

The extension $\mathbb{Q}_3(\sqrt{-3})/\mathbb{Q}_3$ is ramified of degree two, hence tame. Its character χ_{-3} is nontrivial on inertia and trivial on wild inertia. On the plus space all inertia operators are unipotent. Choose an inertia element on which $\chi_{-3} = -1$; its action on the minus space has all eigenvalues -1, so it fixes no nonzero vector. Consequently

$$(W_{4,\ell}^{-})^{I_3} = 0, \qquad P_3^{-}(T) = 1.$$

Both spaces are tame, so both Swan conductors vanish. The Artin conductor formula [27] is

$$a_3(V) = \dim V - \dim V^{I_3} + \mathrm{Sw}_3(V).$$

It gives $a_3(W^{+}) = 3$, $a_3(W^{-}) = 4$, completing Theorem 8.1.

9 The local factor and conductor exponent at 2

Let $\ell \neq 2$ and retain the rational descents defined by the literal cubic deck operation and inversion. On $T = \mathbb{P}^1_z \setminus \{0, \infty, \mu_3\}$ let $f : \mathcal{E} \to T$ be the family $y^2 = x(x-1)(x-z^3)$, and put

$$\mathcal{H}_\ell = H^1_!(T_{\overline{\mathbb{Q}}}, \mathrm{Sym}^4 R^1 f_* \mathbb{Q}_\ell), \qquad W_{4,\ell}^{\pm} = (e_{\mathrm{nt}} \mathcal{H}_\ell)^{I=\pm 1}.$$

The following argument uses one arithmetic Galois cover for the complete calculation. The semistable-model criterion supplies tameness; the main, tail and node terms then determine the inertia invariants.

Theorem 9.1 (Dyadic local data)**.** *For every coefficient prime $\ell \neq 2$,*

$$\mathcal{H}_\ell^{I_2} = 0, \qquad \mathrm{Sw}_2(\mathcal{H}_\ell) = 0.$$

Consequently both specified rank-four descents satisfy

$$d_{2,\ell}^{\pm} = 0, \qquad P_{2,\ell}^{\pm}(T) = 1, \qquad \mathrm{Sw}_2(W_{4,\ell}^{\pm}) = 0, \qquad a_2(W_{4,\ell}^{\pm}) = 4.$$

The proof occupies the remainder of this section. It retains the arithmetic action on the cover and on its coefficient system separately. The identity $W_{4,\ell}^{-} = W_{4,\ell}^{+} \otimes \chi_{-3}$ is consistent with the equal dyadic data: χ_{-3} is unramified at 2 and has Frobenius value -1 there.

9.1 The exact Galois closure and its arithmetic action

Fix an auxiliary prime $\ell \neq 2$, put $K_0 = \mathbb{Q}_2(\zeta)$, where $\zeta^3 = 1 \neq \zeta$, and let $L = K_0(\pi)$ with $\pi^3 = 2$. Choose an arithmetic inertia lift σ of the generator of $\mathrm{Gal}(L/K_0)$ with $\sigma(\pi) = \zeta\pi$. The following computations concern actual geometric deck transformations and this arithmetic automorphism separately.

9.1.1 The constant in the cubic closure

Set $h = v^3$, $D = 2h + 1$, and $z = v(h + 2)/D$. The polynomial

$$f_z(X) = X^4 - 2zX^3 + 2X - z$$

factors after adjoining v as

$$f_z(X) = (X - v)\left(X^3 - \frac{3v}{D}X^2 - \frac{3v^2}{D}X + \frac{h+2}{D}\right).$$

Adjoin η satisfying

$$\eta^3 = 4(v^3 + 1). \tag{30}$$

The other three roots are exactly

$$v_i = \frac{v - \zeta^i\eta^2/2 - \zeta^{-i}v^2\eta}{D}, \qquad i = 0, 1, 2. \tag{31}$$

Indeed, after the substitution $X = Y + v/D$, the cubic has coefficients

$$p = -\frac{6v^2(h+1)}{D^2}, \qquad q = \frac{2(h+1)(2h^2+h+1)}{D^3}.$$

The two Cardano terms $u = -\eta^2/(2D)$ and $w = -v^2\eta/D$ satisfy $uw = -p/3$ and $u^3 + w^3 = -q$. The discriminant is

$$\operatorname{disc}(f_z) = -432(z^3 - 1)^2 = -3\big(12(z^3 - 1)\big)^2.$$

Consequently $K_0(v, \eta)$ is the splitting field over $K_0(z)$: it contains the four roots and has degree $4 \cdot 3 = 12$, since $v^3 + 1$ has simple zeros and is not a cube in $K_0(v)$. Its group is A_4. Over $\mathbb{Q}_2(z)$ the group is S_4 and the constant extension is exactly K_0. For example, the smooth cubic has the rational point $(v, \eta) = (-1, 0)$, so its function field is geometrically integral and introduces no additional constants.

The smooth projective closure is an elliptic curve, with the displayed point as origin. Explicitly,

$$x = \frac{3\eta}{v+1}, \qquad y = \frac{9(1-v)}{v+1} \quad \Longrightarrow \quad y^2 = x^3 - 27.$$

The normal Klein four subgroup of A_4 consists of translations by the four two-torsion points of this elliptic curve. The quotient generator is

$$c : (v, \eta) \longmapsto (v, \zeta\eta).$$

One can verify the translation assertion directly: translation by $(3, 0)$ sends x to $3(x + 6)/(x - 3)$ and y to $-27y/(x-3)^2$; the resulting v is v_0 in (31). Conjugation by c gives the other two nontrivial translations.

9.1.2 Why the deck action is the projective three-torsion action

For the Legendre family with $\lambda = z^3$, put

$$\alpha(v) = \frac{(v^3 + 2)^2}{3(2v^3 + 1)}.$$

Direct substitution gives

$$3\alpha^4 - 4(1 + \lambda)\alpha^3 + 6\lambda\alpha^2 - \lambda^2 = 0.$$

Thus α is the x-coordinate of a nonzero point of order three, or equivalently one of the four lines in the two-dimensional $\mathbf{F}_3$-space of three-torsion. The rational map $v \mapsto \alpha$ has degree six, whereas

$v \mapsto z$ has degree four. Therefore $\alpha \notin \overline{\mathbb{Q}}_2(z)$: otherwise the degree six would be a multiple of four. The subgroup C_3 fixing v is maximal in A_4, so this already implies

$$\overline{\mathbb{Q}}_2(z, \alpha) = \overline{\mathbb{Q}}_2(v).$$

The normal closures are equal. Hence the A_4 cover just constructed is precisely the splitting cover of the four projective three-torsion lines, with their faithful permutation action. Quadratic twists do not change these lines and act trivially on Sym^4.

9.2 The semistable marked model of the true splitting cover

Use the splitting cover and notation of the preceding subsection:

$$\eta^3 = 4(v^3+1), \qquad e = \eta/\pi^2, \qquad e^3 = v^3+1, \qquad \pi^3 = 2,$$

over $L = \mathbb{Q}_2(\zeta_3, \pi)$. Write $h = v^3$. For completeness, the coordinate changes are explicit. Put

$$q = \frac{v^3+2}{v^3-1}, \qquad \lambda = z^3 = \frac{q^3(q+2)}{(q-1)(q+1)^3}, \qquad \alpha = \frac{q^2}{q^2-1} = \frac{(v^3+2)^2}{3(2v^3+1)}.$$

After quadratically twisting the Legendre curve by α, the rational three-torsion point and its tangent slope are

$$X_0 = \alpha^2, \qquad Y_0 = \frac{q^5}{(q-1)^3(q+1)^4}, \qquad M_0 = \frac{q(q^2+q+1)}{(q-1)(q+1)^2}.$$

Translation by this flex gives $y^2 + 2M_0xy + 2Y_0y = x^3$. Scale first by $q/((q-1)(q+1))$ and then by π. The resulting equation is

$$y^2 + \pi^2 A(v)xy + B(v)y = x^3, \qquad A = \frac{3(h^2+h+1)}{(h-1)(2h+1)}, \quad B = \frac{(h+2)^2}{(h-1)(2h+1)}. \tag{32}$$

The quadratic twist disappears on $\mathrm{Sym}^4 R^1$. Throughout this section, scaling by k means $x = k^2x'$, $y = k^3y'$, so that (a_1, a_3) changes to $(a_1/k, a_3/k^3)$.

The discriminant before pulling back to e is

$$\Delta = \frac{3^6 h^2(h+1)^2(h+2)^6}{(h-1)^6(2h+1)^6}.$$

On the v-line, the fourteen cusp fibres are I_6 at $v = 0, \infty$ and at the three roots of each of $h = -2$, $2h+1 = 0$, and $h = 1$; the three fibres at $h = -1$ are I_2. At either pole divisor $h = 1$ or $2h+1 = 0$, scaling by the inverse of its local parameter makes a_1 a unit, a_3 vanish to order two, and the discriminant have order six. Thus these poles do not conceal additive fibres. The cubic cover $e^3 = h+1$ ramifies at the three I_2 points, changing them to I_6, and is unramified at all other cusps. Consequently all 36 cusp fibres on the true splitting cover are I_6.

The projective Fermat curve $C : e^3 = v^3 + 1$ has good reduction at 2. Its marked cusp sections form exactly nine clusters in the special fibre: three over $v = 0$, three at infinity, and three over $e = 0$. Use respectively v, $w = 1/v$, and e as local parameters. In each cluster the four sections have parameter zero and three parameters whose cubes are ± 2. The complementary coordinate is a unique power series by Hensel's lemma, its derivative being three times a unit. Blow up the nine disjoint ideals $(\text{parameter}, \pi)$. The resulting proper regular model $\mathcal{B}/\mathcal{O}_L$ has one smooth genus-one main component and nine rational tails, all of multiplicity one. Every tail has four disjoint cusp sections and meets the main component at a fifth point. All 36 sections meet the smooth locus of the special fibre and avoid the nine nodes.

This is the stable model of the curve with its full cusp divisor: each rational tail has five special points. Uniqueness of the stable marked model extends the deck group $G = A_4$ and the semilinear arithmetic action σ to $\mathcal{B}$. It also shows that the explicit blowups do not depend on choices of labels within a cusp orbit.

9.2.1 Good models over the nine nodes

The following equations are obtained directly from (32). Their denominators are units in the indicated node neighbourhoods.

At each node over zero put $vr = \pi$ and scale by $k = v^2$. Then

$$a_1 = r^2 A(v), \qquad a_3 = \frac{(1+r^3)^2}{(v^3-1)(2v^3+1)}.$$

At each infinity node put $w = 1/v$, $wr = \pi$, and scale by $k = 1/w$. The coefficients become

$$a_1 = \frac{3r^2(1+w^3+w^6)}{(1-w^3)(1+r^3)}, \qquad a_3 = \frac{(1+2w^3)^2}{(1-w^3)(1+r^3)}.$$

At each node over $e = 0$ put $er = \pi$, $h = e^3 - 1$, and scale by $k = 1/e$. Since $h - 1 = e^3 - 2 = e^3(1 - r^3)$, the coefficients are

$$a_1 = \frac{3r^2(h^2+h+1)}{(1-r^3)(2h+1)}, \qquad a_3 = \frac{(h+2)^2}{(1-r^3)(2h+1)}.$$

At the closed point of every node they reduce to $(a_1, a_3) = (0, 1)$. The Tate discriminant $a_3^3(a_1^3 - 27a_3)$ is therefore a unit. These are smooth elliptic models over the node rings themselves, not only good models over the generic points of the components.

9.2.2 Integral lifts on all three kinds of tails

Write $U = u^3$. At a zero tail put $v = \pi u$ and scale (32) by $k = \pi^2$. One obtains

$$a_1 = \frac{3(4U^2+2U+1)}{(2U-1)(4U+1)}, \qquad a_3 = \frac{(U+1)^2}{(2U-1)(4U+1)}.$$

At an infinity tail put $w = \pi u$ and scale by $k = 1/(\pi(U+1))$. The integral lift is

$$a_1 = \frac{3(1+2U+4U^2)}{1-2U}, \qquad a_3 = \frac{(1+4U)^2(U+1)^2}{1-2U}.$$

At an $e = 0$ tail put $e = \pi u$, so $h = 2U - 1$, and scale by $k = 1/(\pi(U-1))$. This gives

$$a_1 = \frac{3(h^2+h+1)}{2h+1}, \qquad a_3 = \frac{(U-1)^2(h+2)^2}{2h+1}.$$

Every displayed denominator is a unit along its finite special tail. All three lifts reduce to the same ordinary elliptic family

$$E_u : \quad y^2 + xy + (u^3+1)^2 y = x^3, \qquad \Delta = u^6(u^3+1)^6. \tag{33}$$

It has four I_6 fibres, at $u = 0$ and the three points $u^3 = 1$; their a_1 is a unit. At the attaching point $u = \infty$, scaling by u^2 gives good reduction, agreeing with the node charts.

On the main component, away from its nodes, scale (32) by v^2/e. Its reduction is the constant supersingular curve $E_0 : y^2 + y = x^3$. The node charts extend this good reduction at all attaching points. The nine tails account for exactly $9 \cdot 4 = 36$ fibres of type I_6, agreeing with the full generic cusp divisor. Thus no horizontal collision or extra discriminant component remains. The ratios of the main scale to the three node scales are $1/e$, $1/(we)$, and v^2. The ratios of the three tail scales to their node scales are u^{-2}, $u/(u^3+1)$, and $u/(u^3-1)$. All are units on their respective overlaps; the latter two node charts exclude the vanishing of their displayed denominators. These explicit coordinate transitions glue the Weierstrass line bundle and cubic models. At a horizontal cusp, c_4 is a unit and the exact discriminant divisor is $\Delta = \text{unit} \cdot t^6$, with t a local equation of its section. The nodal deformation is therefore etale locally $xy = \text{unit} \cdot t^6$; the unit can be absorbed in one coordinate. The resulting coefficient system has good reduction at every vertical node and multiplicative reduction precisely along the 36 disjoint horizontal sections. The functorial proper semistable/logarithmic cohomology construction is consequently equivariant for G and σ.

9.3 From the charts to arithmetic tameness

Lemma 9.2 (Tameness from a semistable marked elliptic model)**.** *Let K be a finite extension of $\mathbf{Q}_p$, let L/K be a finite tamely ramified extension, and fix a prime $\ell \neq p$. Let U/K be a smooth geometrically connected curve, let $f : E \to U$ be an elliptic scheme, and put $\mathcal{V} = R^1 f_* \mathbf{Q}_\ell$. Suppose that there is a finite étale morphism $g : U' \to U_L$ of degree $d > 0$ and an elliptic scheme $f' : E' \to U'$ with*

$$\operatorname{Sym}^4 R^1 f'_* \mathbf{Q}_\ell \simeq g^* \operatorname{Sym}^4 \mathcal{V}.$$

Suppose in addition that U' has a proper semistable marked model $(\mathcal{C}, \mathcal{D})/\mathcal{O}_L$ with the following properties.

1. *The reduced horizontal divisor $\mathcal{D}$ is a disjoint union of étale sections in the smooth locus of $\mathcal{C}$, and $U' = \mathcal{C}_L \setminus \mathcal{D}_L$.*
2. *The elliptic scheme E' extends to a proper flat family $\mathcal{E} \to \mathcal{C}$ which is smooth over $\mathcal{C} \setminus \mathcal{D}$. In particular it is smooth at every node of the vertical fibre of $\mathcal{C}$.*
3. *Along $\mathcal{D}$ the family is semistable multiplicative: étale locally its singularities have equations $xy = t^m$ for some $m \geq 1$, where $t = 0$ defines a section of $\mathcal{D}$. An invertible factor on the right is allowed.*

Then wild inertia acts trivially on $H^1(U_{\overline{K}}, \operatorname{Sym}^4 \mathcal{V})$, and hence on its parabolic subspace

$$H^1_!(U_{\overline{K}}, \operatorname{Sym}^4 \mathcal{V}) := \operatorname{im}\big(H^1_c(U_{\overline{K}}, \operatorname{Sym}^4 \mathcal{V}) \longrightarrow H^1(U_{\overline{K}}, \operatorname{Sym}^4 \mathcal{V})\big).$$

The same conclusion holds for every Galois-stable direct summand or subquotient of these representations.

Proof. Give $S = \operatorname{Spec} \mathcal{O}_L$ its standard divisorial log structure and $\mathcal{C}$ the log structure defined by its reduced vertical fibre together with $\mathcal{D}$. This is a fine and saturated log-smooth curve over S. Give $\mathcal{E}$ the log structure defined by the inverse image of this boundary, taking its fine and saturated toric model where necessary. The stated node charts show that $\mathcal{E} \to \mathcal{C}$ is log smooth. Here no condition $p \nmid m$ is needed for $xy = t^m$: the toric relation is $e_x + e_y = m e_t$, and the group cokernel relative to $\mathbf{Z}e_t$ is free, generated by $e_x = -e_y$. An invertible factor can be absorbed in x or y. At a vertical base node the elliptic family is smooth, so its log chart is just the smooth pullback of $uv = \varpi$. Thus composition gives a proper fine and saturated log-smooth model over S.

Form the fourfold fibre product in the category of fine and saturated log schemes,

$$\mathcal{X} = \mathcal{E} \times^{\mathrm{fs}}_{\mathcal{C}} \mathcal{E} \times^{\mathrm{fs}}_{\mathcal{C}} \mathcal{E} \times^{\mathrm{fs}}_{\mathcal{C}} \mathcal{E}.$$

Log smoothness is stable under this base change and composition. The underlying scheme of $\mathcal{X}$ is proper over S: its toric saturation is finite over the ordinary proper fibre product. The locus where the log structure is trivial is exactly

$$X^\circ = E' \times_{U'} E' \times_{U'} E' \times_{U'} E'.$$

In particular it is an open subset of the generic fibre. The horizontal-boundary theorem [33, Corollary 8.4.3, p. 310] applies to the trivial-log open of this proper log-smooth fine and saturated model. The original nearby-cycle and comparison statements are [28, Variant 3.4.1, Proposition 4.2 and Remark 4.2.1]. Consequently wild inertia of L is trivial on $H^q(X^\circ_{\overline{L}}, \mathbf{Z}/\ell^n)$ for every q, n, and hence on its $\mathbf{Q}_\ell$ cohomology. One must use the horizontal-boundary statement here; the generic log structure on the proper compactification is not trivial.

For completeness, the required coefficient cohomology is an explicit summand of ordinary degree-five cohomology of X°. Write $F : X^\circ \to U'$ and $\mathcal{V}' = R^1 f'_* \mathbf{Q}_\ell$. Let ι_i be inversion on factor i, and let S_4 permute the four factors. The commuting idempotents

$$e_{\mathrm{odd}} = \prod_{i=1}^{4} \frac{1-\iota_i^*}{2}, \qquad e_{\mathrm{sym}} = \frac{1}{4!} \sum_{\sigma \in S_4} \mathrm{sgn}(\sigma)\sigma^*, \qquad e = e_{\mathrm{odd}} e_{\mathrm{sym}}$$

are defined over L. In relative Künneth cohomology, e_{odd} retains only the tensor of four degree-one factors. Thus it kills $R^b F_* \mathbf{Q}_\ell$ for $b \neq 4$ and has image $(\mathcal{V}')^{\otimes 4}$ for $b = 4$. A transposition of two degree-one factors contributes the Koszul sign -1; the sign in e_{sym} therefore makes its action on this tensor power the ordinary symmetrizer. It follows that

$$eR^b F_* \mathbf{Q}_\ell = \begin{cases} \mathrm{Sym}^4 \mathcal{V}', & b = 4, \\ 0, & b \neq 4. \end{cases}$$

Applying this projector to the Leray spectral sequence leaves only this one row. Its canonical edge identification is

$$eH^5(X^\circ_{\overline{L}}, \mathbf{Q}_\ell) \simeq H^1(U'_{\overline{L}}, \mathrm{Sym}^4 \mathcal{V}').$$

All maps are Galois equivariant, so the right-hand side is tame.

Finally the finite étale pullback and trace maps satisfy

$$\mathrm{Tr}_g \circ g^* = d\,\mathrm{id} \quad \text{on} \quad H^1(U_{\overline{K}}, \mathrm{Sym}^4 \mathcal{V}).$$

After dividing the trace by d, pullback is a split injection into the tame coefficient cohomology just obtained. The parabolic image is a Galois-stable subspace of ordinary cohomology. Since L/K is tamely ramified, its wild inertia subgroup equals the wild inertia subgroup of K under the usual embedding $G_L \subset G_K$. This proves all assertions. □

Corollary 9.3 (Wild inertia on the cubic Legendre group)**.** *For every $\ell \neq 2$, the full parabolic group $\mathcal{H}_\ell$ and both specified descents $W^{\pm}_{4,\ell}$ have zero Swan conductor at 2.*

Proof. Apply Lemma 9.2 to the true splitting cover

$$z = \frac{v(v^3+2)}{2v^3+1}, \qquad e^3 = v^3 + 1, \qquad L = \mathbb{Q}_2(\zeta_3, \pi), \quad \pi^3 = 2.$$

This constant extension has ramification index three and is tame. If $h = v^3$, then

$$z^3 - 1 = \frac{(h-1)^3(h+1)}{(2h+1)^3}, \qquad \frac{dz}{dv} = \frac{2(v^3-1)^2}{(2v^3+1)^2}.$$

The degree-four rational map ramifies only over the original cusps. The Fermat projection ramifies at $v^3 = -1$, which also maps into $z^3 = 1$; it is unramified at the three points above infinity. Removing all preimages of $\{0, \infty, \mu_3\}$ therefore gives a finite etale degree-twelve cover. Its compactification has genus one and its removed divisor has the 36 points already accounted for above. The displayed integral charts and gluing maps verify each model hypothesis of the lemma: separated etale horizontal sections, good elliptic reduction at the nine vertical nodes, and horizontal singularities $xy = \mathrm{unit} \cdot t^6$. The twist parameter $\alpha = (h+2)^2/(3(2h+1))$ is invertible away from these cusps, so it introduces no additional singular locus on the generic open. The fourth symmetric power removes this quadratic twist. The lemma proves tameness of the full ordinary coefficient cohomology, hence of its parabolic subspace and the two rational projected summands. □

9.4 Geometric deck action and arithmetic inertia

9.4.1 The good main component and two different order-three actions

Set

$$e = \eta/\pi^2, \qquad e^3 = v^3 + 1.$$

This Fermat model is smooth and proper over $\mathcal{O}_L$ and has supersingular genus-one special fibre $\overline{C}$. In particular $\overline{C}[2](\overline{\mathbf{F}}_2)$ has just its origin. The translations by two-torsion extend to this good model and specialize to the identity. Thus K_4 fixes the entire main component pointwise. In contrast, both c and σ act on its base by $e \mapsto \zeta e$; for σ this follows from $\sigma(\eta) = \eta$ and $\sigma(\pi^2) = \zeta^2\pi^2$.

Their actions on the elliptic coefficient are different. The exact Tate equation derived above is

$$E^* : y_*^2 + \pi^2 A(v) x_* y_* + B(v) y_* = x_*^3, \quad A = \frac{3(h^2+h+1)}{(h-1)(2h+1)}, \quad B = \frac{(h+2)^2}{(h-1)(2h+1)}.$$

It was obtained by a scale π after transformations defined over $K_0(v)$, up to the harmless quadratic twist. Hence

$$c : (x_*, y_*) \mapsto (x_*, y_*), \qquad \sigma : (x_*, y_*) \mapsto (\zeta x_*, y_*).$$

On the main component use the further scale $k = v^2/e$, and write $X = x_*/k^2$, $Y = y_*/k^3$. Its special equation is the constant curve

$$E_0 : Y^2 + Y = X^3.$$

For instance its a_3 before reduction is $B(h+1)/h^2$, which reduces to one. Transport of coordinates gives the exact special-fibre actions

	$\overline{C}$	E_0
c	$e \mapsto \zeta e$	$(X, Y) \mapsto (\zeta^2 X, Y)$
σ	$e \mapsto \zeta e$	$(X, Y) \mapsto (X, Y)$.

(34)

Possible signs from a choice of a square root in the preliminary quadratic twist disappear on the fourth symmetric power.

The quotient of $\overline{C}$ by $e \mapsto \zeta e$ is the rational v-line. Since three is invertible in the ℓ-adic coefficient field, finite-map trace gives

$$H^1(\overline{C}, \mathbb{Q}_\ell)^\sigma = 0.$$

Thus, by (34), the entire main component term

$$H^1(\overline{C}, \mathbb{Q}_\ell) \otimes \mathrm{Sym}^4 H^1(E_0, \mathbb{Q}_\ell)$$

has no σ-invariants. This remains true for its A_4-invariant part and for every subsequent projector. The conclusion uses both the base and coefficient actions, not the coefficient action alone.

9.4.2 The projective quaternion action on the constant coefficient

The full A_4 action on projective three-torsion remains faithful after reduction: on a good elliptic model, three-torsion is finite etale and its four projective lines remain distinct. Since K_4 fixes the main base pointwise, its action on the constant elliptic coefficient is therefore faithful in $\mathrm{Aut}(E_0)/\{\pm 1\}$. All these are zero-preserving elliptic isomorphisms. Their projective ambiguity is exactly the quadratic-twist sign already removed in Sym^4.

For completeness, the inverse image of this K_4 is the quaternion group Q_8. Its eight automorphisms are

$$(X, Y) \longmapsto (X + r, Y + r^2 X + t), \qquad r \in \mathbf{F}_4, \quad t^2 + t = r^3.$$

For $r \neq 0$ their squares are $[-1] : (X, Y) \mapsto (X, Y + 1)$. The order-three automorphism $(X, Y) \mapsto (\zeta^2 X, Y)$ permutes the three nonidentity elements of the projective K_4. Together these generate the binary tetrahedral group $Q_8 \rtimes C_3$.

Let $M = \mathrm{Sym}^4 H^1(E_0, \mathbb{Q}_\ell)$. The quaternion character calculation gives

$$\dim M^{K_4} = \dim M^{Q_8} = 2, \qquad M^{Q_8 \rtimes C_3} = 0. \tag{35}$$

Indeed the traces on Sym^4 are 5 on the two central elements and 1 on the six elements of order four. On each of the sixteen remaining binary tetrahedral elements the trace is -1. Averaging gives $(10 + 6)/8 = 2$ and $(10 + 6 - 16)/24 = 0$. More precisely, if c_0 is either generator of the quotient C_3, then

$$\mathrm{Tr}(c_0 \mid M^{Q_8}) = \frac{1}{8} \sum_{q \in Q_8} \mathrm{Tr}(c_0 q \mid M) = -1.$$

Together with the absence of fixed vectors this shows that its characteristic polynomial on M^{K_4} is $X^2 + X + 1$, so its eigenvalues are ζ, ζ^2. This calculation is valid for every $\ell \neq 2$, including $\ell = 3$, because coefficients have characteristic zero.

9.4.3 Arithmetic action on the two node orbits

The three attaching points over $z = 0$ are $v = 0$, $e^3 = 1$; the three over $z = \infty$ are the three values of e/v at infinity. In each case c and σ permute these three points in the same cycle. The stabilizer in A_4 is K_4. Use the good node scale $k = v^2$ at zero and $k = 1/w$ at infinity, where $w = 1/v$. These scales contain neither e nor π. Thus c acts trivially on the displayed elliptic node coordinates, whereas σ acts by $(X, Y) \mapsto (\zeta X, Y)$.

Identifying the A_4-invariant direct sum over either three-point orbit with M^{K_4} at one point, the arithmetic action is therefore the coefficient action of $c^{-1}\sigma$. It is the order-three rotation $(X, Y) \mapsto (\zeta X, Y)$. By (35), neither of these two two-dimensional node terms has arithmetic invariants. Their duals and Tate twists have the same property, since the ℓ-adic Tate character is unramified at two.

At each of the remaining three main attaching points $e = 0$, the whole geometric A_4 is the stabilizer. Its projective action is the full tetrahedral action above, so its node coefficient invariant space is zero by (35). The main degree-zero constant coefficient invariant space is zero for the same reason.

Thus every main and node term surviving geometric descent has no arithmetic eigenvalue one. The tail calculation and the exact cohomological passage below complete the invariant calculation.

9.5 The descended ordinary tails

The degree-four tail quotient can be read from the exact rational map. At zero, $v = \pi u$ gives

$$\frac{z}{\pi^4} = \frac{u(u^3 + 1)}{4u^3 + 1} \quad \longmapsto \quad Z = u^4 + u.$$

This is the translation quotient by $K_4 = (\mathbb{F}_4, +)$. Since $\lambda = z^3 = 16Z^3$, the reduced Legendre j-invariant is $1/Z^6$. Infinity is its inversion counterpart.

At a tail over a cube root of unity, use $e = \pi u$ and $h = 2u^3 - 1$. The exact identity

$$\lambda - 1 = \frac{(h - 1)^3 (h + 1)}{(2h + 1)^3}$$

gives

$$T = \frac{\lambda - 1}{16} = \frac{u^3 (u^3 - 1)^3}{(4u^3 - 1)^3} \quad \longmapsto \quad (u^4 + u)^3.$$

This is the degree-twelve quotient by the affine group $A_4 = (\mathbb{F}_4, +) \rtimes \mathbb{F}_4^\times$. Its reduced j-invariant is $1/T^2$. The ordinary characteristic-two family

$$\mathcal{K}_t : \quad y^2 + xy = x^3 + tx$$

has $j = 1/t^2$. Hence the two descended fourth-symmetric-power systems are respectively its cubic pullback $t = Z^3$ and its ordinary version $t = T$. Ordinary elliptic curves with the same j differ by a quadratic twist; this ambiguity disappears on Sym^4. Over the ordinary generic point, the zero-preserving automorphism group is $\{\pm 1\}$. Thus the identification is compatible with the descended fourth-symmetric-power action, independently of the choice of elliptic isomorphism.

These identifications include middle extensions at the missing points. Indeed finite pushforward on the compact quotient curves is exact, and finite-group invariants over $\mathbb{Q}_\ell$ are exact. The invariant pushforward of a middle-extension coefficient system is the middle extension of its generic descent, as may also be checked on the local inertia-invariant stalks. This applies to the wildly ramified special translation quotients as well: it uses characteristic-zero coefficients, not tameness of those finite maps. Thus the component parabolic cohomology after deck descent is exactly that of the displayed ordinary or cubic Kloosterman system. The cited Kloosterman calculation gives zero for both of these parabolic Sym^4 groups, accounting for every descended tail middle term.

9.6 Vanishing of the descended tail terms

Lemma 9.4 (The characteristic-two Kloosterman tails)**.** *Let $k = \overline{\mathbf{F}}_2$, let $\ell \neq 2$, and let $f : E \to \mathbf{G}_{m,k}$ be the elliptic scheme*

$$E_t : \quad y^2 + xy = x^3 + tx.$$

Put $\mathcal{F} = \mathrm{Sym}^4 R^1 f_* \mathbf{Q}_\ell$, and let $[3] : \mathbf{G}_m \to \mathbf{G}_m$ send Z to Z^3. Then

$$H^1_!(\mathbf{G}_m, \mathcal{F}) = 0, \qquad H^1_!(\mathbf{G}_m, [3]^*\mathcal{F}) = 0.$$

For the first sheaf the Swan conductor at infinity and the dimensions of the inertia-invariant spaces at $0, \infty$ are $(1, 1, 0)$; for the second they are $(3, 1, 2)$.

Proof. On the open set $x \neq 0$, put $u = y/x$. The elliptic equation becomes $u^2 + u = x + t/x$, the Artin–Schreier cover defining the rank-two Kloosterman sheaf Kl_2 in characteristic two. The involution $u \mapsto u + 1$ is elliptic inversion. The two omitted points $x = 0$ and the point at infinity are fixed by this involution, so their boundary cohomology is invariant. Taking the odd part of the compact-support exact sequence identifies $R^1 f_* \mathbf{Q}_\ell$ with Kl_2. This also fixes the trace convention: its Frobenius trace is minus the usual Kloosterman sum. The elliptic realization is recorded in Haessig [31], proof of Theorem 2.2, citing Schoof–van der Vlugt, Proposition 3.1(4).

We use the characteristic-two local calculation of Yun [30], §4.3.1 and Lemmas 4.3.3–4.3.4. The image of inertia at infinity on Kl_2 is the binary tetrahedral group $2.A_4$, with lower ramification filtration

$$D_0 = 2.A_4, \quad D_1 = Q_8, \quad D_2 = D_3 = \{\pm 1\}, \quad D_4 = 1.$$

In the fourth symmetric power this gives $\mathrm{Swan}_\infty(\mathcal{F}) = 1$ and $\mathcal{F}^{I_\infty} = 0$. At zero, the local monodromy is a single unipotent block, so $\dim \mathcal{F}^{I_0} = 1$. The global geometric monodromy is SL_2; therefore $H^0(\mathbf{G}_m, \mathcal{F}) = H^2_c(\mathbf{G}_m, \mathcal{F}) = 0$. The Grothendieck–Ogg–Shafarevich formula and the boundary exact sequence consequently give

$$\dim H^1_!(\mathbf{G}_m, \mathcal{F}) = \mathrm{Swan}_\infty(\mathcal{F}) - \dim \mathcal{F}^{I_0} - \dim \mathcal{F}^{I_\infty} = 1 - 1 - 0 = 0.$$

The cubic pullback kills the tame quotient $(2.A_4)/Q_8$ of the local image at infinity, leaving image Q_8. On Sym^4, its invariant plane is generated by $x^4 + y^4$ and x^2y^2. Tame pullback multiplies the Swan conductor by three. Thus the corresponding local data are $\mathrm{Swan}_\infty = 3$, invariant dimension 1 at zero, and invariant dimension 2 at infinity. A finite-index restriction still has geometric Zariski closure SL_2, so the global H^0 and H^2_c again vanish. The same formula now reads $3 - 1 - 2 = 0$. □

Remark 9.5. The lemma applies to the ordinary tail $t = (\lambda - 1)/16$ and to the cubic tails $\lambda = 16Z^3$ supplied by the explicit Legendre coordinate changes, after quadratic twists are removed by Sym^4. The statement concerns parabolic cohomology of these characteristic-two families. Identifying their contributions to the original dyadic inertia invariants still requires the component and node maps of the semistable model; the vanishing alone is not a local-factor calculation.

9.7 Equivariant nearby cycles and parabolic descent

We apply the preceding calculations to the proper marked semistable model $\mathcal{B}/\mathcal{O}_L$ of the actual splitting cover $\eta^3 = 4(v^3 + 1)$. Its main component is $C : e^3 = v^3 + 1$, where $e = \eta/\pi^2$. Write $G = A_4$ for the geometric deck group, $c : e \mapsto \zeta e$ for the indicated geometric order-three operation, and $\sigma(\pi) = \zeta\pi$ for the arithmetic descent operation. The two operations on the coefficient are those computed in the arithmetic-action calculation above; they must not be identified.

The marked model has a genus-one main component and nine rational tails, each with four marked cusp sections. It is a stable marked curve. Consequently the deck and arithmetic descent operations extend to it by uniqueness of the stable marked model. Let $\mathcal{D}$ be the union of the horizontal cusp sections, and write

$$j : \mathcal{B} \setminus \mathcal{D} \hookrightarrow \mathcal{B}, \qquad \mathcal{F} = \mathrm{Sym}^4 R^1 f_* \mathbf{Q}_\ell, \qquad \mathcal{J} = j_* \mathcal{F}.$$

Here f is the good elliptic family on the indicated open, supplied by the integral charts. We assert equivariance for $\mathcal{F}$, not for an arbitrary choice of the preliminary quadratic twist of the elliptic family. On the generic open, $\mathcal{F}$ is canonically the pullback of the original fourth symmetric-power coefficient. Its deck and arithmetic identifications extend to the lisse sheaf on $\mathcal{B} \setminus \mathcal{D}$: restriction of lisse sheaves to a dense open of a normal scheme is fully faithful. They then extend to $\mathcal{J}$ by functoriality of j_*. In particular the following sequences carry the actual G and arithmetic actions.

9.7.1 The local nearby-cycle calculation

Let $\mathcal{B}_a$ denote the normalized special-fibre components, and let $\mathcal{J}_a$ be their coefficient middle extensions across their horizontal cusps. For each of the nine vertical nodes x, put $H_x = \mathcal{F}_x$ and set

$$\mathcal{N} = \bigoplus_{x \text{ a vertical node}} H_x.$$

The good elliptic charts at the nodes are essential here. The semistable-curve calculation gives

$$R^0\Psi(\mathcal{J}_{\bar\eta}) = \mathcal{J}_{\bar s}, \qquad R^1\Psi(\mathcal{J}_{\bar\eta}) = \bigoplus_x (H_x(-1))_x, \qquad R^q\Psi(\mathcal{J}_{\bar\eta}) = 0 \quad (q > 1). \tag{36}$$

At a vertical node the coefficient is lisse on the full ring $uv = \pi$. Thus the assertion is the constant-sheaf node calculation tensored with H_x; see [33, Corollary 8.4.4] for that semistable specialization of the nearby-cycle formula.

There are no extra vertical vanishing cycles at the horizontal cusps. The base is smooth there, the marked section is a relative smooth divisor, and the elliptic family is multiplicative with local parameter $q = t^6$ times a unit. After strict henselization the branches split and the unit has every prime-to-2 root. The coefficient therefore has its usual tame unipotent monodromy in t, and its horizontal invariant sheaf $\mathcal{F}^{I_t}$ is lisse over the trait. More precisely, [32, Expose XIII, Lemmas 2.1.10–2.1.11] gives base change for $Rj_*\mathcal{F}$ and $R\Phi(j_!\mathcal{F}) = 0$ for a lisse sheaf tamely ramified along a relative normal-crossings divisor. Apply also smooth local acyclicity to the invariant sheaf on the section. The exact sequence

$$0 \longrightarrow j_!\mathcal{F} \longrightarrow j_*\mathcal{F} \longrightarrow i_*\mathcal{F}^{I_t} \longrightarrow 0$$

then gives $R\Phi(j_*\mathcal{F}) = 0$ at that section. The statements apply at finite prime-to-2 coefficient levels and then to $\mathbf{Q}_\ell$ coefficients. Away from the cusps and vertical nodes, ordinary smooth

local acyclicity applies. These observations prove (36) and identify the restrictions of $\mathcal{J}_{\bar{s}}$ with the component middle extensions.

Proper base change and the five-term nearby-cycle sequence yield

$$0 \longrightarrow H^1(\mathcal{B}_{\bar{s}}, \mathcal{J}_{\bar{s}}) \longrightarrow H^1(\mathcal{B}_{\bar{\eta}}, \mathcal{J}_{\bar{\eta}}) \longrightarrow \mathcal{N}(-1) \longrightarrow H^2(\mathcal{B}_{\bar{s}}, \mathcal{J}_{\bar{s}}). \tag{37}$$

For the proper comparison, see [32, Expose XIII, §2.1.8, especially (2.1.8.9)]. We only use that the quotient of the second term by the first embeds in $\mathcal{N}(-1)$; surjectivity onto the node module is unnecessary.

For normalization $\nu : \coprod_a \mathcal{B}_a \to \mathcal{B}_{\bar{s}}$, lissity at the nodes gives

$$0 \longrightarrow \mathcal{J}_{\bar{s}} \longrightarrow \nu_* \bigoplus_a \mathcal{J}_a \longrightarrow \bigoplus_x (H_x)_x \longrightarrow 0.$$

The last map is the difference of the two restrictions. Orient every edge from the unique genus-one main component toward its tail. Both G and σ preserve this orientation, so there is no additional sign in their action on the node module. The cohomology sequence gives

$$\bigoplus_a H^0(\mathcal{B}_a, \mathcal{J}_a) \longrightarrow \mathcal{N} \longrightarrow H^1(\mathcal{B}_{\bar{s}}, \mathcal{J}_{\bar{s}}) \longrightarrow \bigoplus_a H^1(\mathcal{B}_a, \mathcal{J}_a) \longrightarrow 0. \tag{38}$$

Both sequences are equivariant. Taking G-invariants is exact over $\mathbf{Q}_\ell$, also when $\ell = 3$.

9.7.2 Finite descent, including the wild special-tail quotients

We recall why taking the indicated stabilizer invariants on a special tail computes its descended middle cohomology even when the quotient is wildly ramified at the attaching point. Let $q : Y \to X = Y/H$ be any of the finite special-tail quotient maps. On the dense open where the coefficient descends, let F_X be the descended sheaf and let j_Y, j_X be the respective open immersions. Finite pushforward and exactness of rational H-invariants give

$$\left(q_* j_{Y*} q^* F_X\right)^H = j_{X*} F_X.$$

This can be checked directly on sections: extension by j_* and finite pushforward commute, and invariant sections on the finite cover are precisely sections downstairs. Hence

$$H^1(Y, j_{Y*} q^* F_X)^H = H^1(X, j_{X*} F_X).$$

No tameness of q at the omitted point is required. In the present case the maps on the open tails are the displayed separable quotients $Z = u^4 + u$ and $T = (u^4 + u)^3$. Their ramification at infinity is thus handled by the middle extension, not discarded. The same argument justifies generic finite Galois descent of parabolic cohomology.

The preceding component and group calculations now supply exactly the following inputs:

1. The main-component cohomology has no σ-eigenvalue 1: σ acts on its Fermat-base H^1 with the two nontrivial cubic characters and acts trivially on the constant elliptic coefficient.

2. All descended tail middle-cohomology groups vanish, by their identification with the ordinary and cubic-pullback Kloosterman families and the stated local-monodromy calculation.

3. $\mathcal{N}^G$ consists of the two two-dimensional node spaces over 0 and ∞. Each has arithmetic eigenvalues ζ, ζ^2; the three other node spaces vanish after G-invariants. The Tate twist (-1) is unramified at 2.

Consequently neither $\mathcal{N}^G$ nor $(\bigoplus_a H^1(\mathcal{B}_a, \mathcal{J}_a))^G$ has eigenvalue 1. Equation (38) gives the same conclusion for $H^1(\mathcal{B}_{\bar{s}}, \mathcal{J}_{\bar{s}})^G$. Equation (37) then gives it for $H^1(\mathcal{B}_{\bar{\eta}}, \mathcal{J}_{\bar{\eta}})^G$. Absence of an eigenvalue passes to subspaces, quotients and extensions, so no degeneration assertion or vanishing of graph monodromy is used.

Finally, H^1 of the coefficient middle extension on a proper smooth curve is canonically its parabolic cohomology. Generic finite Galois descent therefore identifies the last G-invariant group with the full original cubic-Legendre parabolic representation. An inertia-invariant vector would be fixed by the chosen arithmetic inertia lift σ, which has no eigenvalue 1 here. Both specified rank-four summands thus have zero inertia invariants. Combining this with Corollary 9.3 gives, for every $\ell \neq 2$,

$$d^+_{2,\ell} = d^-_{2,\ell} = 0, \qquad P^+_{2,\ell}(T) = P^-_{2,\ell}(T) = 1, \qquad a_2(W^+_{4,\ell}) = a_2(W^-_{4,\ell}) = 4.$$

This proves Theorem 9.1. In particular the arithmetic constant in the splitting cover is retained throughout; replacing $\eta^3 = 4(v^3+1)$ by a geometrically isomorphic equation without tracking that constant would not determine this inertia action.

10 Conductor products and coefficient-prime scope

For a realization V_ℓ and a prime $p \neq \ell$, use

$$P_p(V_\ell, T) = \det(1 - \mathrm{Frob}_p\, T \mid V_\ell^{I_p}), \qquad a_p(V_\ell) = \dim V_\ell - \dim V_\ell^{I_p} + \mathrm{Sw}_p(V_\ell).$$

The local proofs are carried out directly with arbitrary coefficient prime different from the residue characteristic. They give

	d_2	Sw_2	a_2	d_3	Sw_3	a_3
$W^+_{4,\ell}$	0	0	4	1	0	3
$W^-_{4,\ell}$	0	0	4	0	0	4

with the p-column interpreted only for $\ell \neq p$. At 3 the invariant line of the plus descent has geometric Frobenius eigenvalue 3. Thus $P_2^+ = P_2^- = P_3^- = 1$ and $P_3^+ = 1 - 3T$.

For $p \geq 5$ and $\ell \neq p$, good-prime unramifiedness follows by the relative tame-specialization argument of Section 3. To justify its coefficient change explicitly, use the integral sheaf on the full cubic cover before applying rational projectors. At boundary points of both generic and special geometric fibres, the Tate-module description gives prime-to-p unipotent local monodromy; at infinity the additional quadratic twist is also tame. The cubic Kummer cover is tame since $p \neq 3$. Thus every finite ℓ-power coefficient level is relatively tame. The same specialization, duality and parabolic-image comparison then applies, and rational projectors pass to its summands. This uses the local Tate descriptions, not a claim that the order of $\mathrm{GL}_2(\mathbb{Z}/\ell^n)$ is prime to p. Hence, for every $\ell \neq 2, 3$,

$$N^{(\ell)}(W^+_{4,\ell}) = \prod_{p \neq \ell} p^{a_p} = 2^4\, 3^3 = 432, \qquad N^{(\ell)}(W^-_{4,\ell}) = 2^4\, 3^4 = 1296.$$

Equivalently these are the conductor products obtained by choosing an auxiliary coefficient prime away from each residue characteristic. Independence of the displayed bad-prime factors and exponents follows from the explicit calculations, without invoking a general bad-prime compatibility theorem. No assertion about the usual inertia invariants of the p-adic realization at $p = \ell$ is being substituted for an away-from-ℓ Euler factor. Nor does this calculation identify the entire Weil–Deligne representation in all coefficient characteristics.

The numerical conductors formerly proposed for the two recipes therefore agree with the conductor products of the specified descents. This does not establish the functional-equation sign, analytic continuation, or the accuracy of the historical central-value computation.

11 Strong irreducibility from local monodromy

Conventions and inputs. Fix a coefficient prime $\ell \neq 3$. Write $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and τ_ℓ for the character of $\mathbb{Q}_\ell(-1)$, so that $\tau_\ell(\mathrm{Frob}_p) = p$ for geometric Frobenius at $p \neq \ell$. To avoid opposite Hodge–Tate sign conventions, call h a *Tate exponent* when a Hodge–Tate summand is $\mathbb{C}_\ell(-h)$. Thus $\mathbb{Q}_\ell(-1)$ has Tate exponent 1.

The inputs for $V = W_{4,\ell}^+$ are:

1. V has dimension four over $\mathbb{Q}_\ell$, is unramified outside a finite set, and is de Rham at ℓ.
2. Its Tate exponents are $\{0, 0, 5, 5\}$. These follow from the specified smooth projective Scholl realization of Theorem 6.1, its rational projector and the Hodge numbers $h^{5,0} = h^{0,5} = 2$, using de Rham and Hodge–Tate comparison.
3. At 3, inertia is unipotent with a monodromy logarithm N having one Jordan block of length four. With F geometric Frobenius,

$$FNF^{-1} = 3^{-1}N,$$

and the eigenvalues on the successive quotients of $0 \subset \ker N \subset \ker N^2 \subset \ker N^3 \subset \ker N^4 = V$ are $3, 9, 27, 81$, respectively.

The third input is the full arithmetic conclusion of Section 8, not an inference from the conductor alone. The same construction gives $W_{4,\ell}^- = W_{4,\ell}^+ \otimes \chi_{-3}$.

Lemma 11.1 (Rank-one characters over $\mathbb{Q}$)**.** *Let $\delta : G_{\mathbb{Q}} \to \mathbb{Q}_\ell^\times$ be a continuous character, unramified outside a finite set, and Hodge–Tate at ℓ with Tate exponent m. Then $\delta = \epsilon\tau_\ell^m$ for a finite-order character ϵ.*

Proof. Tate's theorem identifies a Hodge–Tate abelian local representation with a locally algebraic one when its inertia action is semisimple. A character automatically satisfies the latter condition. The precise primary reference is Serre [29, Chapter III, §1.2, Theorem (Tate), printed p. III-7; proof in the Appendix to Chapter III]; the definition in §1.1 identifies the local algebraic power. In the present $\mathbb{Q}_\ell$-valued rank-one case, it follows that $\delta\tau_\ell^{-m}$ has finite image on inertia at ℓ.

Put $\epsilon = \delta\tau_\ell^{-m}$. By Kronecker–Weber, an abelian character of $G_{\mathbb{Q}}$ is a continuous character of $\widehat{\mathbb{Z}}^\times = \prod_p \mathbb{Z}_p^\times$. Outside a finite set its restrictions to the unit factors are trivial, by the unramified hypothesis. For each remaining $p \neq \ell$, its restriction to $\mathbb{Z}_p^\times$ has finite image: its open pro-p subgroup maps into $\mathbb{Q}_\ell^\times$, whose compact subgroups have an open pro-ℓ subgroup, so that image is finite. At ℓ finiteness was just proved. Thus the entire image is contained in the product of finitely many finite groups, and is finite. The exponent m is the stated Tate exponent, since finite characters have Tate exponent zero. □

Lemma 11.2 (The single-block flag)**.** *Let k be a field of characteristic zero and let N on k^4 have one nilpotent Jordan block. Its invariant subspaces are exactly $\ker N^r$ for $0 \leq r \leq 4$.*

Proof. As a $k[t]$-module, with t acting as N, this space is $k[t]/(t^4)$. Its submodules are the ideals $(t^j)/(t^4)$, equivalently the displayed kernels. This proof remains valid after arbitrary scalar extension. □

Theorem 11.3. *Under the stated inputs, both $W_{4,\ell}^+$ and $W_{4,\ell}^-$ are absolutely irreducible for every $\ell \neq 3$.*

Proof. It suffices to treat $V = W_{4,\ell}^+$. Suppose $0 \neq U \subsetneq V \otimes_{\mathbb{Q}_\ell} \overline{\mathbb{Q}}_\ell$ is a $G_{\mathbb{Q}}$-stable subspace, of dimension $r \in \{1, 2, 3\}$. Its finitely many basis coordinates belong to a finite coefficient extension,

so there is no issue of defining a continuous representation over an infinite coefficient field. There is in fact a stronger descent argument in this case.

Some inertia element at 3 acts as $\exp(cN)$ with $c \neq 0$. Since its logarithm is a finite polynomial in $\exp(cN) - 1$, stability under that element makes U stable under N. Lemma 11.2 gives

$$U = (\ker N^r) \otimes_{\mathbb{Q}_\ell} \overline{\mathbb{Q}}_\ell.$$

Set $U_0 = \ker N^r \subset V$. This is defined over $\mathbb{Q}_\ell$ and is $G_{\mathbb{Q}}$-stable: extension of scalars makes every $\rho(g)$ preserve it, and the original matrices are $\mathbb{Q}_\ell$-linear. Hence its determinant $\delta = \det(U_0)$ is a $\mathbb{Q}_\ell$-valued character.

The category of de Rham representations is closed under subobjects, quotients and exterior powers; its comparison functor is exact and strict for the filtrations. Thus U_0 and δ are de Rham, the Tate exponents of U_0 form a submultiset of $\{0, 0, 5, 5\}$, and the Tate exponent m of δ is their sum. In particular

$$m \in 5\mathbb{Z}.$$

For clarity, the dimension part of closure under subobjects follows already from left exactness of D_{dR} and $\dim D_{\mathrm{dR}}(M) \leq \dim M$: apply this to U_0, V and V/U_0, with equality for V. The usual strictness of de Rham comparison supplies the asserted filtered submultiset statement.

The character δ is unramified outside a finite set, so Lemma 11.1 gives $\delta = \epsilon\tau_\ell^m$. Since U_0 has unipotent inertia at 3, its determinant is unramified there, as is ϵ. On the other hand the explicitly ordered monodromy flag gives

$$\delta(\mathrm{Frob}_3) = 3^{1+2+\cdots+r} = 3^{r(r+1)/2}.$$

Therefore

$$\epsilon(\mathrm{Frob}_3) = 3^{r(r+1)/2-m}.$$

The left side is a root of unity. A rational integer power of 3 is a root of unity only when its exponent is zero. Consequently $m = r(r+1)/2$. For $r = 1, 2, 3$ these values are $1, 3, 6$, respectively, none of which belongs to $5\mathbb{Z}$. This contradiction proves absolute irreducibility of V. Tensoring with the one-dimensional character χ_{-3} preserves absolute irreducibility and proves the minus case. □

Corollary 11.4 (Restriction to finite extensions)**.** *For every finite extension $L/\mathbb{Q}$, the restrictions of either $W_{4,\ell}^{\pm}$ to G_L are absolutely irreducible, for $\ell \neq 3$.*

Proof. It suffices again to treat the plus representation. If a proper nonzero subspace U is stable under G_L, replace L by a finite Galois extension containing it and put $H = G_L \triangleleft G_{\mathbb{Q}}$. The intersection $H \cap I_3$ has finite index in I_3, hence its image under the nonzero tame monodromy parameter still contains a nonzero element. Consequently every H-stable subspace is N-stable. Normality makes every gU also H-stable, and all have dimension r. By Lemma 11.2, all equal $\ker N^r$ after scalar extension. Thus U is $G_{\mathbb{Q}}$-stable, contradicting Theorem 11.3. □

Limits and consequences. The argument uses the written Scholl/Hodge and local-at-3 results as inputs; it does not re-prove their constructions. It makes no assertion for the coefficient prime $\ell = 3$, since the local monodromy calculation above uses $\ell \neq 3$. No conjecture of semisimplicity is used. Proper induction from a smaller-dimensional representation over a finite extension is excluded by Corollary 11.4. An ordinary tensor product of two rank-two continuous ℓ-adic representations over finite coefficient extensions is also excluded: after a finite local extension its monodromy operator would have form $N_1 \otimes 1 + 1 \otimes N_2$, with $N_i^2 = 0$, hence its cube would be zero, whereas the actual N^3 is nonzero. These exclusions are independent of the dyadic calculation in Section 9 and do not establish automorphy or determine the Sato–Tate group.

12 Current claims and their dependencies

"Written proof" below describes the arguments in this manuscript. It does not mean external peer review or formal verification. The coefficient-prime restrictions are part of the conclusions.

Conclusion	Proof location and main inputs	Scope or remaining limit
Ranks and rational descents	Sections 3, 5: parabolic dimension, literal inversion and deck projectors	Specified arithmetic normalization; $W_4^- = W_4^+ \otimes \chi_{-3}$.
Pairing and good Euler factors	Sections 5, 7: cup product, exact finite sums and inert boundary term	Exact formulas at $p \geq 5$; no certification of every stored numerical row.
Good-prime weight five	Corollary 5.8: Deligne's weight theorem and specialization	Independent of empirical reciprocal-quartic tests.
Rational Chow realization and Hodge numbers	Theorem 6.1: full-level cover, finite projectors, Scholl comparison	All ℓ; $h^{5,0} = h^{0,5} = 2$; no other Hodge types.
Local data at 3	Section 8: quotient family, collision cohomology, ordered Frobenius flag	Every $\ell \neq 3$; plus $P_3 = 1 - 3T$, $a_3 = 3$; minus $P_3 = 1$, $a_3 = 4$.
Local data at 2	Section 9: exact Galois closure, semistable charts, tame comparison, tails and nodes	Every $\ell \neq 2$; both $d_2 = 0$, $P_2 = 1$, $\mathrm{Sw}_2 = 0$, $a_2 = 4$.
Conductor products	Section 10: both local calculations and good-prime unramifiedness	Plus 432, minus 1296; for $\ell \neq 2, 3$, these are the prime-to-ℓ Artin conductors.
Strong absolute irreducibility	Section 11: single-block monodromy at 3, Hodge–Tate exponents and rank-one characters over $\mathbb{Q}$	Every $\ell \neq 3$, after every finite number-field extension; coefficient prime 3 is not covered.
Cubic second-moment comparison	Section 4: geometric trace/determinant, stated comparison theorem, seven exact rows and Sturm bound 24	Separate rank-two object; bounded exact certificate retained; orders $4, 6$ do not inherit this proof.

The local factor and exponent independence asserted here is obtained directly for all allowable auxiliary primes. A full comparison of the bad-prime Weil–Deligne representations, including a treatment at equal coefficient and residue characteristic, is a different statement. The remaining research questions are listed in Section 13.

Sign dictionary. At an inert good prime, the literal plus descent $I = +1$ has trace $-S_p - 1$ and conductor 432. The literal minus descent $I = -1$ has trace $S_p + 1$ and conductor 1296. Thus an older numerical label "plus sign" for the finite-sum formula means the latter object; it is not the literal plus eigenspace of I.

Review priorities. The most useful independent review targets are the arithmetic constant in the dyadic closure, the coefficient action on the main component and nodes, the horizontal-boundary nearby-cycle hypotheses, and the local-at-3 global-to-local map with its Frobenius normalization. The strong-irreducibility deduction depends on the complete local flag, not merely on its conductor exponent. These are review targets within the stated proof, not newly claimed external endorsements.

13 Numerical evidence, corrections and open questions

Theorem 1.1 identifies the *exact good-prime recipe*. It does not turn each stored approximation to that recipe into exact data. The code supplement preserves the earlier correction package and

its logs unchanged. No archived research program, prime scan, certificate generator or numerical L-function evaluator was executed for this consolidation. The previously supplied standalone implementation rechecked only the bounded cubic certificate and Sturm comparison in Section 4; its results are retained separately in the supplied supplement. The local continuations used only the small exact algebra checks recorded in their supplements. No archived point-counting scan or numerical L-function experiment has been rerun for this consolidation.

13.1 Evidence retained from the earlier computations

The carried record contains direct exact small-prime counts, an exact finite modular-comparison certificate, rounded FFT-derived larger tables, and high-precision evaluations using those tables. These are different evidence classes. An exact consistency check on rounded input does not certify how the input was obtained. The proof of the reciprocal template also does not validate every historical value of its first two coefficients.

The historical 1296 model imposed $P_2(T) = P_3(T) = 1$. The later 432 twist in the retained N14/N15 record used $P_2(T) = 1$ and $P_3(T) = 1 - 3T$. The blanket statement in version 2 that both experiments imposed $P_2 = P_3 = 1$ was therefore too broad. The new local proofs establish these factors and conductor products for W_4^- and W_4^+, respectively; they do not use the numerical fits. The historical phrase "plus-sign recipe" refers to inert trace $S_p + 1$, which is the *literal minus descent* W_4^- of this manuscript. For that 1296 model the recorded functional-equation sign is $+1$ and the central value is approximately 2.2413552. Those two statements remain numerical observations, not consequences of the local proofs. The candidate completion is

$$\Lambda(s) = N^{s/2}\Gamma_{\mathbb{C}}(s)^2\mathcal{L}(s), \qquad \Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s).$$

The associated theta-kernel Mellin transform used in the code has the normalization $\widehat{\Lambda} = \Lambda/4$. Working precision and small finite-sum defects do not supply an interval enclosure or an infinite-tail bound. The historical gamma-factor shape agrees with the independent Hodge calculation in Section 6. The conductor is now independently established by the local arguments. The gamma-factor agreement and conductor calculation do not prove the functional-equation sign or analytic continuation.

Likewise, a symmetrized evaluator

$$\sum_{n \le N} a_n\{G(s, n, t_0) + \epsilon G(6 - s, n, 1/t_0)\}$$

has built-in symmetry at $t_0 = 1$. For $s = 3$ and $\epsilon = -1$ it vanishes for any coefficient sequence. This is not an independent analytic-rank test. The historical N18 failure caused by comparing an approximately fifty-digit reference against a seventy-digit requirement is retained as a failure; other reported agreements are not promoted to certified analytic digits.

13.2 Two exact algebraic checks and their limits

Lemma 13.1 (Reciprocal-quartic modulus criterion)**.** *Let $P > 0$, $c_1, c_2 \in \mathbb{R}$, $D = c_1^2 - 4(c_2 - 2P)$ and $B = 16P - c_1^2 - D$. Every root of $X^4 - c_1X^3 + c_2X^2 - Pc_1X + P^2$ has modulus $\sqrt{P}$ if and only if*

$$D \ge 0, \qquad B \ge 0, \qquad B^2 \ge 4c_1^2D.$$

Proof. Put $y = X + P/X$. The two y-values are the roots of $y^2 - c_1y + c_2 - 2P$. The required condition is that they are real and both lie in $[-2\sqrt{P}, 2\sqrt{P}]$, equivalently $D \ge 0$ and $|c_1| + \sqrt{D} \le 4\sqrt{P}$. Squaring with the nonnegative signs retained gives the stated test. □

This repairs the earlier floor-square-root validator; it tests specified coefficients. Good-prime purity of the exact geometric factors follows independently from Corollary 5.8; it does not certify any stored coefficient table.

If the same quartic factors as $(X^2 - aX + P)(X^2 - bX + P)$, coefficient comparison gives $D = (a-b)^2$. A nonsquare rational D excludes this model with $a, b \in \mathbb{Q}$. If a, b are conjugate in a fixed quadratic field, every nonzero D has that field's square class. Neither this limited obstruction nor empirical trace distributions establishes general irreducibility, rules out every inner twist, or determines the Sato–Tate group. The independent proof in Section 11 now establishes strong absolute irreducibility for $\ell \neq 3$; it does not rely on this test.

13.3 Remaining mathematical questions

The bad-prime factors and conductor exponents have been determined in Sections 8–10. Their auxiliary-prime scope is recorded there; they are no longer open obligations of this draft. The following questions remain separate.

1. The strong-irreducibility argument uses $\ell \neq 3$. Its extension at coefficient prime 3, a full bad-prime Weil–Deligne comparison, and precise Sato–Tate or inner-twist classifications are not established here.

2. Identify any automorphic representation with all required hypotheses checked. The manuscript does not establish quaternion multiplication or transfer an adjacent automorphy theorem.

3. Establish analytic continuation and the functional equation by a theorem, including its sign. A root-number calculation requires its own local epsilon-factor inputs; the conductor alone does not determine it.

4. Fix rational period structures before interpreting critical values or determinant ratios arithmetically. The retained numerical central values are not certified analytic evaluations.

5. For character orders $4, 6$ in the older second-moment work, supply their separate geometric inputs. Their finite comparison rows do not follow from the cubic-only proof as all-prime theorems.

These are limits of the present conclusions, not instructions to pursue all of them before this manuscript can be independently reviewed. The rank-four object is not the rank-two H^1 of an elliptic curve, even after a Tate twist. No comparison controlling the Mordell–Weil rank or BSD leading coefficient of a specified elliptic curve has been established here. A future motivic special-value formulation would require its own precise twist, arithmetic groups and comparison theorem.

A Known quadratic trace identities

Let χ be the quadratic character of $\mathbb{F}_p$, extended by zero, and put $T_j(p) = \sum_{\lambda \neq 0,1} \chi(\lambda) a_p(\lambda)^j$. The following is a known-trace-formula corollary.

Theorem A.1. *For every prime $p \geq 5$,*

$$T_4(p) = -c(p) - b(p) - 3p\,a(p) - 2p^2 - 3p - 1,$$

where $a = 8.4.a.a = \eta(2z)^4\eta(4z)^4$, $b = 4.6.a.a = \eta(2z)^{12}$, and $c = 8.6.a.a = q + 20q^3 - 74q^5 - 24q^7 + \cdots$.

Proof. Use the Ahlgren and Ahlgren–Ono trace formulas as stated in [16, Theorem 2.3]:

$$-\operatorname{Tr}_{k+2}(\Gamma_0(4),p) = 3 + \sum_{\lambda\neq 0,1} G_{k+2}(a(\lambda),p),$$
$$-\operatorname{Tr}_{k+2}(\Gamma_0(8),p) = 4 + \sum_{\gamma\neq 0,1,-1} G_{k+2}(a(\gamma^2),p).$$

Here $G_4(s,p) = s^2 - p$ and $G_6(s,p) = s^4 - 3ps^2 + p^2$. Put $U_k = \sum_{\lambda\neq 0,1} a(\lambda)^k$ and $Q_k = \sum_{\gamma\neq 0,1,-1} a(\gamma^2)^k$. Then $Q_k - U_k = T_k$, and $Q_0 - U_0 = -1$. The relevant spaces have bases

$$S_4(\Gamma_0(4)) = 0, \quad S_4(\Gamma_0(8)) = \langle a\rangle, \quad S_6(\Gamma_0(4)) = \langle b\rangle,$$
$$S_6(\Gamma_0(8)) = \langle b(z), b(2z), c(z)\rangle.$$

Subtracting the weight-four trace formulas gives $-a(p) = 1 + T_2(p) + p$, so $T_2 = -a(p) - p - 1$. Subtracting the weight-six formulas gives $-(b(p)+c(p)) = 1 + T_4 - 3pT_2 - p^2$. Substitution proves the formula. □

The identity is a consequence of established trace formulas. No new modularity or priority assertion is attached to this appendix.

B The corrected hypergeometric normalization

Assume $p \geq 5$, $p \equiv 1 \pmod 3$. Let ϕ and ρ be the quadratic and a primitive cubic character, and let g and J denote Gauss and Jacobi sums for a fixed nontrivial additive character. We use the normalized Gauss-sum definition [16, Eq. (1.1)], with all multiplicative characters, including the trivial character, extended by zero at zero. Put

$$H(\lambda) = \frac{1}{1-p}\sum_{\chi} c(\chi)\chi(\lambda), \qquad c(\chi) = \frac{g(\phi\chi)^2 g(\bar\chi)^2}{g(\phi)^2}.$$

Only nonzero arguments are used here; the separately defined value at zero is immaterial. For nonsingular λ, $a(\lambda) = \phi(-1)H(\lambda)$. Fix a generator ω of the multiplicative character group such that $\rho = \omega^{(p-1)/3}$ and $\phi = \omega^{(p-1)/2}$. The same ω is used to interpret the rational parameters of $H_p(A;1)$. Let

$$A = \left\{\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{3}; 1, 1, \tfrac{5}{6}, \tfrac{5}{6}\right\}, \qquad J = J(\rho, \phi\bar\rho).$$

Lemma B.1. *With this normalization,*

$$M_{3,2}(p) + 1 = \frac{1}{p-1}\sum_{\chi} c(\chi)c(\bar\rho\bar\chi) = -J^2 H_p(A;1). \tag{39}$$

Proof. Since ϕ is nontrivial, $c(\chi) = J(\phi\chi,\bar\chi)^2$ by the Gauss–Jacobi identity. Expanding the Jacobi sums and summing over χ imposes $xy = (1-x)(1-y)$, equivalently $x + y = 1$. Therefore

$$\sum_{\chi} c(\chi) = (p-1)\sum_{x} \phi(x(1-x)) = -(p-1)\phi(-1),$$

so $H(1) = \phi(-1)$. Character orthogonality applied to $H(\lambda)^2$ then gives the first equality of (39).

The numerator of each summand for $H_p(A;1)$ is

$$g(\phi\chi)^2 g(\rho\chi)^2 g(\bar\chi)^2 g(\phi\bar\rho\bar\chi)^2.$$

Its normalizing denominator is $g(\phi)^2 g(\rho)^2 g(\varepsilon)^2 g(\phi\bar\rho)^2$, where $g(\varepsilon)^2 = 1$. The convolution on the middle of (39) has denominator $g(\phi)^4$ and the opposite overall sign, since $1 - p = -(p-1)$. Their ratio is

$$-\left(\frac{g(\rho)g(\phi\bar\rho)}{g(\phi)}\right)^2 = -J^2.$$

This proves the normalization factor. □

For instance, choose $\omega(3) = e^{2\pi i/6}$ at $p = 7$ and $\rho = \omega^2$. Then $J = 2 + \sqrt{-3}$, $J^2 = 1 + 4\sqrt{-3}$, and $M_{3,2}(7) + 1 = -15$. Hence $H_7(A; 1) = 15/(1 + 4\sqrt{-3})$, which is not -15. The v1 phrase identifying the convolution directly with the normalized A-value must therefore include the factor in (39). The underlying moment calculation is unchanged.

Under the carried-certificate comparison of Section 4, the corrected modular relation is

$$J^2 H_p(A; 1) = a_f(p) + p.$$

For $B = \{\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}; 1, 1, 1, 1\}$, the published identity $H_p(B; 1) = a_f(p) + \left(\frac{3}{p}\right) p$ [12, Theorem 2 and Table 1] gives

$$J^2 H_p(A; 1) = H_p(B; 1) + \left(1 - \left(\frac{3}{p}\right)\right) p.$$

The datum A also occurs in the EHMM literature at a different argument [19]. A priority claim requires comparison of both the argument and the Jacobi twists. The algebra above supplies the normalization proof; numerical scripts are retained separately as historical checks.

C An analytic period substitution and the row-sign correction

This appendix records a change of variables and a sign convention. Relations explicitly marked numerical refer to the historical candidate model and are not critical-value theorems.

C.1 An exact change of variables

Let

$$F(\lambda) = {}_2F_1\left(\tfrac{1}{2}, \tfrac{1}{2}; 1; \lambda\right), \qquad F_c(\lambda) = F(1 - \lambda), \qquad 0 < \lambda < 1.$$

For an integer $k \geq 0$, an integer $0 \leq j \leq k$, and a real number $e > 0$, define $y = F_c/F$. The inverse parameter is the modular lambda function at iy. Put

$$G_e(iy) = \lambda(iy)^e (1 - \lambda(iy)) \theta_3(0 \mid iy)^{2k+4}.$$

Then the following analytic identity holds:

$$\int_0^1 \lambda^{e-1} F(\lambda)^j F_c(\lambda)^{k-j} \, d\lambda = \pi \int_0^\infty G_e(iy) y^{k-j} \, dy. \tag{40}$$

Proof. Both $u = F$ and $v = F_c$ solve

$$\lambda(1 - \lambda)u'' + (1 - 2\lambda)u' - \tfrac{1}{4}u = 0.$$

This is the specialization of the hypergeometric differential equation in DLMF 15.10.1 [20]. Abel's identity therefore gives

$$uv' - u'v = \frac{C}{\lambda(1 - \lambda)}.$$

The hypergeometric–elliptic identity and the logarithmic endpoint expansion (DLMF 19.5.1 and 19.12.1) yield

$$F(\lambda) = 1 + O(\lambda), \qquad F_c(\lambda) = \frac{1}{\pi} \log \frac{16}{\lambda} + O(\lambda |\log \lambda|) \quad (\lambda \longrightarrow 0^+).$$

The differentiated expansions imply $C = -1/\pi$. Hence

$$\frac{dy}{d\lambda} = -\frac{1}{\pi \lambda (1 - \lambda) F(\lambda)^2}. \tag{41}$$

The positive function y decreases from $+\infty$ to 0. DLMF 20.9.2, together with DLMF 19.5.1, identifies $F(\lambda(iy)) = \theta_3(0 \mid iy)^2$ and $F_c(\lambda(iy))/F(\lambda(iy)) = y$. Using $F_c = yF$ and $d\lambda = -\pi\lambda(1-\lambda)F^2\,dy$ proves (40) after reversing the endpoints. The original integrand is $O(\lambda^{e-1}(1+|\log\lambda|)^{k-j})$ near 0 and $O((1+|\log(1-\lambda)|)^j)$ near 1. Both bounds are integrable under the stated hypotheses, so the improper change of variables is justified. □

For the present candidate periods, $k = 4$ and $e \in \{1/3, 2/3, 4/3, 5/3\}$; writing $m = k + 1 - j$ gives the factor y^{m-1} used in the old note N12. The same argument also covers its $k = 2$ control integrals. This establishes the analytic dictionary and its normalization. Identifying its right-hand side with a specified arithmetic L-value requires further modularity, rational-structure, and Mellin-normalization arguments. In particular, the rational Scholl realization in Section 6 does not by itself identify these integrals with rational period bases or prove the rational determinant constants below.

C.2 The documented and implemented signs differ

Write $H_j(\lambda) = F(\lambda)^j F_c(\lambda)^{4-j}$ and

$$C_c(j) = \int_0^1 (\lambda^c + \lambda^{-c}) H_j(\lambda)\,d\lambda.$$

For two row families $A_{2/3}, A_{1/3}$ use the fixed row order

$$\mu_A(j_1, j_2) = \det \begin{pmatrix} A_{2/3}(j_1) & A_{2/3}(j_2) \\ A_{1/3}(j_1) & A_{1/3}(j_2) \end{pmatrix}.$$

N12, line 45, describes the uniform convention

$$S_c^{\mathrm{unif}}(j) = \int_0^1 (\lambda^c - \lambda^{-c}) H_j(\lambda)\,d\lambda.$$

The actual N11 implementation instead uses

$$S_{2/3}^{\mathrm{code}}(j) = \int_0^1 (\lambda^{-2/3} - \lambda^{2/3}) H_j(\lambda)\,d\lambda,$$
$$S_{1/3}^{\mathrm{code}}(j) = \int_0^1 (\lambda^{1/3} - \lambda^{-1/3}) H_j(\lambda)\,d\lambda.$$

Thus, exactly and without a numerical computation,

$$\mu_{S^{\mathrm{code}}}(j_1, j_2) = -\mu_{S^{\mathrm{unif}}}(j_1, j_2).$$

The old negative ratios use the implemented convention. To retain the uniform convention in prose, the corresponding *reported numerical* relations must instead read

$$\frac{\mu_{S^{\mathrm{unif}}}(1,3)}{\mu_C(2,4)} \overset{\mathrm{num}}{=} \frac{1}{2}, \qquad \frac{\mu_{S^{\mathrm{unif}}}(0,2)}{\mu_C(1,3)} \overset{\mathrm{num}}{=} \frac{4}{3},$$
$$\mathcal{L}_{\mathrm{candidate}}(4) \overset{\mathrm{num}}{=} \frac{\pi^6 \mu_{S^{\mathrm{unif}}}(0,2)}{209952}.$$

Alternatively, explicitly define the two implemented row orientations and retain the negative signs in the historical outputs. No numerical rerun is needed to diagnose this discrepancy. The five principal critical-value relations expressed only through μ_C are unaffected.

Revision record and computational supplement

The public predecessors are the original preprint [1] and its version-2 consolidation [26]. This revision integrates the local arguments, conductor deductions and irreducibility argument into one manuscript. The earlier versions remain historical records; their identifiers do not identify the text of this revision.

The earlier corrections repaired the quartic finite test set, hypergeometric normalization and numerical validators. The cubic proof supplied the geometric inputs for character order three. The good-prime purity and rational full-level-12 Scholl construction supplied the weight, projectors and Hodge numbers. This revision retains those arguments and adds:

- the explicit local-at-3 quotient, inertia calculation and Frobenius sign for the literal plus descent;
- the true arithmetic Galois closure at 2, its semistable model, a full tameness argument, the tail vanishing and the equivariant nearby-cycle calculation;
- the resulting conductor products and their auxiliary-prime scope;
- strong absolute irreducibility for $\ell \neq 3$, with its precise local and Hodge–Tate inputs;
- the current claim/dependency map in Section 12.

The true dyadic splitting cover is $\eta^3 = 4(v^3 + 1)$. The earlier non-Galois cover $s^3 = v^3 - 1$ was a valid tameness device; it was not the arithmetic Galois closure needed for the later inertia calculation. Here the complete true-cover charts also meet the tameness lemma's hypotheses, so the local proof uses that one cover throughout.

The package contains the current TeX sources, current claim/status files, small exact local-check scripts with their retained run reports, and the unchanged earlier source/supplement archive. That historical archive contains the bounded cubic certificate and its independent exact implementation, along with older numerical records. Its earlier open-task lists and metadata describe their original stage and are superseded by the current claim map, not silently rewritten. Source hashes establish file preservation, not mathematical truth. The finite algebra checks do not implement nearby cycles or certify the geometric proof. The numerical L-function experiments and large scans have not been rerun for this consolidation. Both earlier success and failure records remain historical evidence.

AI assistance and review status

The earlier project used ChatGPT and Claude, as disclosed in v1. AI assistance during preparation and revision extended throughout the manuscript, including the abstract, exposition, mathematical arguments and appendices. Its uses included drafting and revising mathematical arguments, preparing code and algebraic and sign checks, looking up references, and preparing and revising the LaTeX source. In particular, the new local, conductor and irreducibility arguments in Sections 8–11 were drafted and internally checked with ChatGPT assistance. This internal AI-assisted checking does not constitute independent specialist review. Model versions are not specified because they are not verified in the available preparation record. AI systems are not authors. Responsibility for the arguments, references, computational record and disclosure rests with the author. No external specialist review, journal acceptance, proof-assistant verification, or exhaustive novelty determination is represented by this manuscript. The newly written local and irreducibility proofs do not establish rank-four automorphy, analytic continuation, the functional-equation sign, arithmetic period identities, or a classical elliptic BSD theorem.

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